Solution of differential equations

Contents

Linearisation of non-linear differential equations

Confine our attention to 2$^{\text{nd}}$ order ODEs
Consider

$$y'' + 3yy' + y^3 = 0$$

under appropriate constraints on domain, linear equations have two linearly independent solutions

$$Ly = 0 \qquad L \text{ } 2^{\text{nd}} \text{ order linear operator}$$

$$\{y = y_1(x), y_2(x)\}$$

$$y'' + xy = 0$$

$$\begin{aligned} y_1(0) = 1, \quad y_2(0) = 0, \\ y_1'(0) = 0, \quad y_2'(0) = 1 \end{aligned}$$

$$y_1(x) \qquad y_2(x)$$

$$y(x) = a_1 y_1(x) + a_2 y_2(x)$$

Two equations are equivalent if they possess the same Lie algebra of symmetries i.e. there exists a point transformation which transforms one equation to the other.

Point transformation

$$x, y \longmapsto X, Y, \qquad X = F(x, y); \quad Y = G(x, y)$$

other transforms depend upon the derivatives e.g. contact

$$x, y \longmapsto X, Y, \qquad X = F(x, y, y'); \quad Y = G(x, y, y')$$

Lie algebra

$\{G_i, \ i = 1, \ldots, n\}$ forms a Lie algebra if

$$[G_i, G_j] = C_{ijk} G_k$$

where the C_{ijk} are constants (called structure constants) and [,] is a skew symmetric operator

$$[G_i, G_j] = G_i G_j - G_j G_i$$

so that it is obvious that

$$[G_i, G_j] = -[G_j, G_i]$$

Note: the structure constants are invariant under a point transformation. i.e. the algebra is independent of the co-ordinate representation being used.

In general a symmetry exists if a "system" is invariant under the transformation induced by the symmetry. An infinitessimal point transformation of x, y to $\bar{x}, \bar{y}$ is represented by

$$\bar{x} = x + \varepsilon\xi(x, y), \qquad \bar{y} = y + \varepsilon\eta(x, y),$$

where ε is the infinitesimal parameter. For $f(x, y)$ an infinitesimal point transformation produces

$$\begin{aligned} f(\bar{x}, \bar{y}) &= f\Big(x + \varepsilon\xi, y + \varepsilon\eta\Big) \\ &= f(x, y) + \varepsilon\left(\xi\frac{\partial f}{\partial x} + \eta\frac{\partial f}{\partial y}\right) + O(\varepsilon^2) \\ &= f(x, y) + \varepsilon G f. \end{aligned}$$

where

$$G = \xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y}$$

is the generator of the infinitesimal point transformation.

$$\left\{G_i = \xi_i\frac{\partial}{\partial x} + \eta_i\frac{\partial}{\partial y}, \quad i = 1, 2, \ldots, n\right\}$$

constitutes a Lie algebra if

$$\begin{aligned} [G_i, G_j] &= \left[\xi_i\frac{\partial}{\partial x} + \eta_i\frac{\partial}{\partial y}, \xi_j\frac{\partial}{\partial x} + \eta_j\frac{\partial}{\partial y}\right] \\ &= \left(\xi_i\frac{\partial\xi_j}{\partial x} + \eta_i\frac{\partial\xi_j}{\partial y} - \xi_j\frac{\partial\xi_i}{\partial x} - \eta_j\frac{\partial\xi_i}{\partial y}\right)\frac{\partial}{\partial x} \\ &\quad + \left(\xi_i\frac{\partial\eta_j}{\partial x} + \eta_i\frac{\partial\eta_j}{\partial y} - \xi_j\frac{\partial\eta_i}{\partial x} - \eta_j\frac{\partial\eta_i}{\partial y}\right)\frac{\partial}{\partial y} \\ &= C_{ijk}\left(\xi_k\frac{\partial}{\partial x} + \eta_k\frac{\partial}{\partial y}\right). \end{aligned}$$

How to deal with $f(x, y, y')$ and $f(x, y, y', y'')$

How do $y', y'', \ldots$ transform?

$$\begin{aligned}
\frac{d\bar{y}}{d\bar{x}} &= \frac{d(y+\varepsilon\eta)}{d(x+\varepsilon\xi)} \\
&= \frac{\dfrac{dy}{dx}+\varepsilon\dfrac{d\eta}{dx}}{1+\varepsilon\dfrac{d\xi}{dx}} \\
&= \frac{y'+\varepsilon\eta'}{1+\varepsilon\xi'} \\
&= y'+\varepsilon(\eta'-y'\xi') \qquad \text{to } 1^{\text{st}} \text{ order in } \varepsilon.
\end{aligned}$$

$$\begin{aligned}
\frac{d^2\bar{y}}{d\bar{x}^2} &= \frac{d}{d\bar{x}}\left(\frac{d\bar{y}}{d\bar{x}}\right) \\
&= \frac{d\left[y'+\varepsilon(\eta'-y'\xi')\right]}{d(x+\varepsilon\xi)} \\
&= \frac{y''+\varepsilon(\eta''-y''\xi'-y'\xi'')}{1+\varepsilon\xi'} \\
&= y''+\varepsilon(\eta''-2y''\xi'-y'\xi'').
\end{aligned}$$

Notes

1. The process may be extended to higher order derivatives. There is a recursion relation. If

$$G = \xi\frac{\partial}{\partial x}+\eta\frac{\partial}{\partial y},$$

then

$$G^{(1)} = G+(\eta'-y'\xi')\frac{\partial}{\partial y'},$$

$$G^{(2)} = G^{(1)}+(\eta''-2y''\xi'-y'\xi'')\frac{\partial}{\partial y''}.$$

Let

$$H^{(1)} = \eta'-y'\xi'.$$

Then

$$G^{(1)} = G+H^{(1)}\frac{\partial}{\partial y'},$$

$$G^{(n)} = G^{(n-1)}+H^{(n)}\frac{\partial}{\partial y^{(n)}}.$$

$$\begin{aligned}
H^{(2)} &= (H^{(1)})'-y''\xi', \\
H^{(3)} &= (H^{(2)})'-y'''\xi'. \\
&\;\;\vdots
\end{aligned}$$

2. The process may be extended to systems of equations with independent variables and dependent variables.

Symmetries of a 2^{nd} order ODE

$$N(x, y, y', y'') = 0,$$

$$G = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y},$$

with

$$\xi = \xi(x, y), \qquad \eta = \eta(x, y)$$

(i.e. point transformations), is the generator of a symmetry (i) if

$$G^{(2)} N = 0 \qquad \text{when } N = 0,$$

where

$$G^{(2)} = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} + (\eta' - y'\xi') \frac{\partial}{\partial y'} + (\eta'' - 2y''\xi' - y'\xi'') \frac{\partial}{\partial y''}.$$

Example: $y'' = 0$

$$G^{(2)} y'' = \eta'' - 2y''\xi' - y'\xi''$$

$G^{(2)} y'' = 0:$

$$\frac{\partial^2 \eta}{\partial x^2} + 2y' \frac{\partial^2 \eta}{\partial x \partial y} + (y')^2 \frac{\partial^2 \eta}{\partial y^2} + y'' \frac{\partial \eta}{\partial y} - y' \left(\frac{\partial^2 \xi}{\partial x^2} + 2y' \frac{\partial^2 \xi}{\partial x \partial y} + (y')^2 \frac{\partial^2 \xi}{\partial y^2} + y'' \frac{\partial \xi}{\partial y} \right) = 0.$$

Since $\xi(x, y)$ and $\eta(x, y)$ are independent of y', equate coefficients of independent powers of y' to zero.

$$(y')^3 : -\frac{\partial^2 \xi}{\partial y^2} = 0 \qquad \text{(ii)}$$

$$(y')^2 : \frac{\partial^2 \eta}{\partial y^2} - 2\frac{\partial^2 \xi}{\partial x \partial y} = 0 \quad \text{(iii)}$$

$$y' : 2\frac{\partial^2 \eta}{\partial x \partial y} - \frac{\partial^2 \xi}{\partial x^2} = 0 \quad \text{(iv)}$$

$$(y')^0 : \frac{\partial^2 \eta}{\partial x^2} = 0 \qquad \text{(v)}$$

$$\text{(ii)} \qquad \xi = a(x)y + b(x).$$

$$\text{(iii)} \qquad \frac{\partial^2 \eta}{\partial y^2} = 2a',$$

$$\eta = a'y^2 + c(x)y + d(x).$$

$$\text{(iv)} \qquad 4a''y + 2c' - a''y - b'' = 0. \qquad \text{(vi)}$$

$$\text{(v)} \qquad a'''y^2 + c''y + d'' = 0. \qquad \text{(vii)}$$

In (vi) and (vii) equate coefficients of independent powers of y to zero:

$$\begin{aligned} y^2 &: a''' = 0, \\ y^1 &: c'' = 0, \\ & \quad a'' = 0, \\ y^0 &: d'' = 0, \\ & \quad b'' = 2c'. \end{aligned}$$

$$\begin{aligned} a &= A_0 + A_1 x, \\ b &= B_0 + B_1 x + C_1 x^2, \\ c &= C_0 + C_1 x, \\ d &= D_0 + D_1 x. \end{aligned}$$

$$\begin{aligned} \xi &= (A_0 + A_1 x)y + B_0 + B_1 x + C_1 x^2, \\ \eta &= A_1 y^2 + (C_0 + C_1 x)y + D_0 + D_1 x. \end{aligned}$$

Observations:

1. There are 8 generators which leave $y'' = 0$ invariant.

2. To obtain all of the symmetries it is necessary to be able to solve the original differential equation.

$G^{(2)}N = 0$ is always linear in ξ and η.

For linear equations in general:

1. There are 8 symmetries.

2. The symmetries close under commutation and have the Lie algebra $\mathrm{sl}(3, R)$.

3. Every linear equation may be transformed into every other linear equation by means of a point transformation.

[Algebras are invariant under point transformations]

4. The transformation is found by comparing the generators of the two equations. More specifically one looks for a subset.

For the free particle

Generator	**Coefficient**
$G_1 = \dfrac{\partial}{\partial x}$	B_0
$G_2 = x\dfrac{\partial}{\partial x}$	B_1
$G_3 = x^2\dfrac{\partial}{\partial x} + xy\dfrac{\partial}{\partial y}$	C_1
$G_4 = y\dfrac{\partial}{\partial x}$	A_0
$G_5 = xy\dfrac{\partial}{\partial x} + y^2\dfrac{\partial}{\partial y}$	A_1
$G_6 = \dfrac{\partial}{\partial y}$	D_0
$G_7 = x\dfrac{\partial}{\partial y}$	D_1
$G_8 = y\dfrac{\partial}{\partial y}$	C_0

$[G_i, G_j]$	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8
G_1	0	G_1	$2G_2 + G_8$	0	G_4	0	G_6	0
G_2		0	G_3	$-G_4$	0	0	G_7	0
G_3			0	$-G_5$	0	$-G_6$	$-G_3$	0
G_4				0	0	$-G_1$	$-G_2$	$-G_4$
G_5					0	$-G_2 - 2G_8$	$-G_3$	$-G_5$
G_6						0	0	G_6
G_7							0	G_7
G_8								0

$$\begin{aligned}
[G_1, G_2] &= \left[\frac{\partial}{\partial x}, x\frac{\partial}{\partial x}\right] \\
&= \frac{\partial}{\partial x}\left(x\frac{\partial}{\partial x}\right) - x\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}\right) \\
&= \frac{\partial}{\partial x} + x\frac{\partial^2}{\partial x^2} - x\frac{\partial^2}{\partial x^2} \\
&= G_1.
\end{aligned}$$

Relationship between symmetries and first integrals

If $I = I(x, y, y')$ has the property that

$$\left.\frac{dI}{dx}\right|_{N(x,y,y',y'')=0} = 0,$$

then $I(x, y, y')$ is a first integral of

$$N(x, y, y', y'') = 0.$$

Suppose G is a symmetry of $N(x, y, y', y'') = 0$. Then the first integral associated with G is $I(x, y, y')$:

$$G^{(1)}I(x, y, y') = 0 \quad \Longrightarrow \quad dI\Big(G^{(1)}\Big) = 0 \quad \text{for surface } N = 0$$

$$\Longrightarrow \left.\frac{dI}{dx}\right|_{N=0} = 0.$$

For $y'' = 0$, consider

$$G_7 = x\frac{\partial}{\partial y}.$$

$$\begin{aligned}
G &= \xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y}, \\
G^{(1)} &= \xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y} + (\eta' - y'\xi')\frac{\partial}{\partial y'}.
\end{aligned}$$

$$G_7^{(1)} = x\frac{\partial}{\partial y} + \frac{\partial}{\partial y'}.$$

$$G_7^{(1)} I = 0\frac{\partial I}{\partial x} + x\frac{\partial I}{\partial y} + \frac{\partial I}{\partial y'} = 0.$$

The equations for the characteristics are

$$\frac{dx}{0} = \frac{dy}{x} = \frac{dy'}{1}.$$

1st and 2nd:

$$\begin{aligned} x\,dx - 0\,dy &= 0, \\ \frac{1}{2}x^2 &= \text{constant}, \\ u &= x \qquad \left(\text{allowed to take any function of } \frac{1}{2}x^2\right). \end{aligned}$$

2nd and 3rd:

$$\begin{aligned} dy - x\,dy' &= 0, \\ dy - u\,dy' &= 0, \\ v = y - uy' &= y - xy'. \end{aligned}$$

i.e.

$$I(u, v) = I(x, y - xy').$$

Imposing the second condition, i.e.

$$\left.\frac{dI}{dx}\right|_{N=0} = 0,$$

$$\frac{\partial I}{\partial u}u' + \frac{\partial I}{\partial v}v' = 0,$$

for which the characteristic is found from

$$\frac{du}{u'} = \frac{dv}{v'}.$$

$$\begin{aligned} \frac{dv}{du} &= \frac{v'}{u'} \\ &= \frac{y' - xy'' - y'}{1} = 0, \qquad v = \text{constant}. \end{aligned}$$

$$I = f(y - xy').$$

usually write

$$I = y - xy'.$$

$$G_3 = x^2\frac{\partial}{\partial x} + xy\frac{\partial}{\partial y}.$$

$$G_3^{(1)}I = x^2\frac{\partial I}{\partial x} + xy\frac{\partial I}{\partial y} + (y - xy')\frac{\partial I}{\partial y'} = 0.$$

Equations for the characteristics are

$$\frac{dx}{x^2} = \frac{dy}{xy} = \frac{dy'}{y - xy'}.$$

1st and 2nd:

$$\frac{dx}{x^2} = \frac{dy}{xy} \quad \Longrightarrow \quad \frac{dy}{y} - \frac{dx}{x} = 0.$$

$$(\log y - \log x = \log u)\,,$$

$$u = \frac{y}{x}.$$

2nd and 3rd:

$$\frac{dy}{xy} = \frac{dy'}{y - xy'}.$$

$$x = \frac{y}{u}.$$

$$\frac{dy}{y^2/u} = \frac{dy'}{y - yy'/u}.$$

$$\frac{dy'}{dy} = \frac{y - \dfrac{yy'}{u}}{y^2/u} = \frac{u - y'}{y}.$$

$$\frac{dy}{y} = \frac{dy'}{u - y'}.$$

$$\begin{aligned} v &= yy' - uy \\ &= yy' - \frac{y^2}{x}. \end{aligned}$$

$$I(u, v) = I\left(\frac{y}{x}, yy' - \frac{y^2}{x}\right).$$

from

$$\frac{\partial I}{\partial u} u' + \frac{\partial I}{\partial v} v' = 0,$$

$$\frac{dv}{du} = \frac{v'}{u'} = \frac{(y')^2 + yy'' - \dfrac{2y'y}{x} + \dfrac{y^2}{x^2}}{\dfrac{y'}{x} - \dfrac{y}{x^2}}.$$

$$
\begin{aligned}
&= \frac{\left(y' - \dfrac{y}{x}\right)^2}{\dfrac{1}{x}\left(y' - \dfrac{y}{x}\right)} \\
&= x\left(y' - \frac{y}{x}\right) \\
&= x\frac{v}{y} \\
&= \frac{v}{u}.
\end{aligned}
$$

$$
\frac{dv}{du} = \frac{v}{u} \quad \Longrightarrow \quad w = \frac{v}{u}.
$$

$$
\begin{aligned}
I &= f\left(\frac{v}{u}\right) \\
&= f\left(\frac{yy' - y^2/x}{y/x}\right) \\
&= f(xy' - y).
\end{aligned}
$$

$$
G_1 = \frac{\partial}{\partial x}.
$$

$$
G_1^{(1)} I = \frac{\partial I}{\partial x} + 0\frac{\partial I}{\partial y} + 0\frac{\partial I}{\partial y'} = 0.
$$

$$
\frac{dx}{1} = \frac{dy}{0} = \frac{dy'}{0}.
$$

1 and 2:

$$
dy - 0\,dx = 0, \qquad u = y.
$$

1 and 3:

$$
dy' - 0\,dx = 0, \qquad v = y'.
$$

$$
\frac{dv}{du} = \frac{v'}{u'} = \frac{y''}{y'} = 0.
$$

$$
w = v.
$$

$$
I = f(v) = f(y').
$$

other form is

$$
\begin{aligned}
I &= f\left(\frac{xy' - y}{y'}\right) \\
&= f\left(x - \frac{y}{y'}\right).
\end{aligned}
$$

$$
I = x - \frac{y}{y'}
$$

is called a resonance integral.

Characteristic structure of a resonance integral is

$$I = a(x) + \frac{b(xy)}{y' - c(x,y)}.$$

Consider

$$y'' + 3yy' + y^3 = 0.$$

(Comes from the study of the modified Emden equation

$$\ddot{q} + \alpha(t)\dot{q} + q^n = 0,$$

which arises in the study of polytropes, i.e. gaseous spheres (models of stars)).

Let

$$G = \xi(x,y)\frac{\partial}{\partial x} + \eta(x,y)\frac{\partial}{\partial y}.$$

Then

$$G^{(2)}\left(y'' + 3yy' + y^3 = 0\right)$$

is

$$(\eta'' - 2y''\xi' - y'\xi'') + 3\eta y' + 3y(\eta' - y'\xi') + 3y^2\eta = 0.$$

$$\begin{aligned}
&\frac{\partial^2\eta}{\partial x^2} + 2y'\frac{\partial^2\eta}{\partial x\partial y} + (y')^2\frac{\partial^2\eta}{\partial y^2} - (3yy' + y^3)\frac{\partial\eta}{\partial y} \\
&\quad + 2(3yy' + y^3)\left(\frac{\partial\xi}{\partial x} + y'\frac{\partial\xi}{\partial y}\right) \\
&\quad - y'\left\{\frac{\partial^2\xi}{\partial x^2} + 2y'\frac{\partial^2\xi}{\partial x\partial y} + (y')^2\frac{\partial^2\xi}{\partial y^2} - (3yy' + y^3)\frac{\partial\xi}{\partial y}\right\} \\
&\quad + 3y\left\{\frac{\partial\eta}{\partial x} + y'\frac{\partial\eta}{\partial y} - y'\frac{\partial\xi}{\partial x} - (y')^2\frac{\partial\xi}{\partial y}\right\} + 3y^2\eta + 3\eta y' = 0.
\end{aligned}$$

Equate coefficients of independent powers of y' to zero:

$$\begin{aligned}
(y')^3 &: \quad -\frac{\partial^2\xi}{\partial y^2} = 0 && \text{(i)} \\
(y')^2 &: \quad \frac{\partial^2\eta}{\partial y^2} + 6y\frac{\partial\xi}{\partial y} - 2\frac{\partial^2\xi}{\partial x\partial y} + 3y/\frac{\partial\xi}{\partial y} - 3y/\frac{\partial\xi}{\partial y} = 0 && \text{(ii)} \\
y' &: \quad 2\frac{\partial^2\eta}{\partial x\partial y} - 3y/\frac{\partial\eta}{\partial y} + 6y/\frac{\partial\xi}{\partial x} + 3y^3\frac{\partial\xi}{\partial y} - \frac{\partial^2\xi}{\partial x^2} + y^3/\frac{\partial\xi}{\partial y} + 3y/\frac{\partial\eta}{\partial y} + 3y\frac{\partial\xi}{\partial x} + 3\eta = 0 && \text{(iii)} \\
(y')^0 &: \quad \frac{\partial^2\eta}{\partial x^2} - y^3\frac{\partial\eta}{\partial y} + 2y^3\frac{\partial\xi}{\partial x} + 3y\frac{\partial\eta}{\partial x} + 3y^2\eta = 0 && \text{(iv)}
\end{aligned}$$

(i)

$$\xi = a(x)y + b(x).$$

(ii)

$$\frac{\partial^2 \eta}{\partial y^2} = -6ay + 2a',$$

$$\eta = -ay^3 + a'y^2 + c(x)y + d(x).$$

(iii)

$$\begin{aligned}&2(-3a'y^2 + 2a''y + c') + 3y^3a - a''y - b'' + 3a'y^2 + 3b'y - 3ay^3 \\ &\qquad + 3a'y^2 + 3cy + 3d = 0.\end{aligned}$$

$$\begin{aligned}y^1 :&\qquad 3a'' + 3b' + 3c = 0, \\ y^0 :&\qquad 2c' - b'' + 3d = 0.\end{aligned}$$

(iv)

$$\begin{aligned}&-a''y^3 + a'''y^2 + c''y + d'' - y^3(-3ay^2 + 2a'y + c) + 2y^3(a'y + b') \\ &\qquad + 3y(-a'y^3 + a''y^2 + c'y + d') + 3y^2(-ay^3 + a'y^2 + cy + d) = 0.\end{aligned}$$

$$\begin{aligned}y^3 :&\qquad -a'' - c + 2b' + 3a'' + 3c = 0, \\ y^2 :&\qquad a''' + 3c' + 3d = 0, \\ y^1 :&\qquad c'' + 3d' = 0, \\ y^0 :&\qquad d'' = 0.\end{aligned}$$

$$\begin{aligned}a'' + b' + c &= 0 \quad (1) \\ b'' - 2c' - 3d &= 0 \quad (2) \\ a''' + 3c' + 3d &= 0 \quad (3) \\ c'' + 3d' &= 0 \quad (4) \\ d'' &= 0 \quad (5)\end{aligned}$$

(5)

$$d = D_0 + D_1 x.$$

(4)

$$c = C_0 + C_1 x - \frac{3}{2} D_1 x^2.$$

(3)

$$a = A_0 + A_1 x + A_2 x^2 - \frac{1}{2}(C_1 + D_0)x^3 + \frac{1}{4} D_1 x^4.$$

(2)

$$b = B_0 + B_1 x + \frac{1}{2}(2C_1 + 3D_0)x^2 - \frac{1}{2} D_1 x^3.$$

check in (1)

$$2A_2 - 3(C_1 + D_0)x + 3D_1x^2 + B_1 + (2C_1 + 3D_0)x - \frac{3}{2}D_1x^2$$
$$+ C_0 + C_1x - \frac{3}{2}D_1x^2 = 0.$$

$$2A_2 + B_1 + C_0 = 0; \qquad -3C_1 - 3D_0 + 2C_1 + 3D_0 + C_1 = 0.$$

$$3D_1 - \frac{3}{2}D_1 - \frac{3}{2}D_1 = 0.$$

i.e. there are 8 independent constants and there are 8 symmetries.

$$G = (ay + b)\frac{\partial}{\partial x} + (-ay^3 + a'y^2 + cy + d)\frac{\partial}{\partial y},$$

$$G_1 = \frac{1}{2}x^2y\frac{\partial}{\partial x} + \left(-\frac{1}{2}x^2y^3 + xy^2 - y\right)\frac{\partial}{\partial y},$$

$$G_2 = y\frac{\partial}{\partial x} - y^3\frac{\partial}{\partial y},$$

$$G_3 = xy\frac{\partial}{\partial x} + (-xy^3 + y^2)\frac{\partial}{\partial y},$$

$$G_4 = \left(-\frac{1}{2}x^2y + x\right)\frac{\partial}{\partial x} + \left(\frac{1}{2}x^2y^3 - xy^2\right)\frac{\partial}{\partial y},$$

$$G_5 = \left(-\frac{1}{4}x^4y + \frac{1}{3}x^3\right)\frac{\partial}{\partial x} + \left(\frac{1}{4}x^4y^3 - x^3y^2 + \frac{3}{2}x^2y - x\right)\frac{\partial}{\partial y},$$

$$G_6 = \left(-\frac{1}{2}x^3y + x^2\right)\frac{\partial}{\partial x} + \left(\frac{1}{2}x^3y^3 - \frac{3}{2}x^2y^2 + xy\right)\frac{\partial}{\partial y},$$

$$G_7 = \left(-\frac{1}{2}x^3y + x\right)\frac{\partial}{\partial x} + \left(\frac{1}{2}x^3y^3 - \frac{3}{2}x^2y^2 + 1\right)\frac{\partial}{\partial y},$$

$$G_8 = -\frac{\partial}{\partial x}.$$

$$y'' = 0.$$

$$I_1 = y',$$

$$I_2 = y'x - y,$$

$$I_3 = x - \frac{y}{y'}.$$

What generators are associated with I_1?

$$G^{(1)}I_1 = 0.$$

$$\left(\xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y} + (\eta' - y'\xi')\frac{\partial}{\partial y'}\right)y' = 0.$$

$$\eta' - y'\xi' = 0.$$

$$\frac{\partial \eta}{\partial x} + y'\frac{\partial \eta}{\partial y} - y'\frac{\partial \xi}{\partial x} - (y')^2\frac{\partial \xi}{\partial y} = 0.$$

$$(y')^2: \qquad \frac{\partial \xi}{\partial y} = 0 \qquad \text{(i)}$$

$$(y')^1: \qquad \frac{\partial \eta}{\partial y} - \frac{\partial \xi}{\partial x} = 0 \qquad \text{(ii)}$$

$$(y')^0: \qquad \frac{\partial \eta}{\partial x} = 0 \qquad \text{(iii)}$$

- **(i)** $\xi = a(x)$.
- **(ii)** $\eta = a'y + b(x)$.
- **(iii)** $a'' = 0$, $b' = 0$.

$$a = A_0 + A_1 x, \qquad b = B_0.$$

The generators

$$G_1 = \frac{\partial}{\partial x}, \qquad A_0,$$

$$G_2 = \frac{\partial}{\partial y}, \qquad B_0,$$

$$G_3 = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}, \qquad A_1.$$

$$[G_1, G_2] = 0,$$

$$[G_1, G_3] = G_1,$$

$$[G_2, G_3] = G_2.$$

We wish to identify the complete symmetry group for the non-linear system (2.22). The generators of (2.22) given in the previous section should have commutation relations appropriate to one of the eight-parameter groups $SL(3,R)$, $SU(3)$ or $GL(2,C)$. The latter two being complex.

The commutation relations are given in Table 2. These relations are appropriate to the symmetry group $SL(3,R)$. Therefore the complete symmetry group for (2.22) is $SL(3,R)$. This will become apparent later.

Table 2

The entry in row i and column j is the Lie bracket $[G_i, G_j]$.

	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8
G_1	0	$-G_2$	$-G_3$	0	G_5	0	$G_7 - 2G_6$	G_3
G_2		0	0	G_2	$G_1 + G_4$	G_3	$3G_3 + G_8$	0
G_3			0	0	G_6	0	$2G_1 - G_4$	G_2
G_4				0	G_5	G_6	$2G_6$	$-(G_3 + G_8)$
G_5					0	0	0	$3G_6 - G_7$
G_6						0	G_5	$2G_4 - G_1$
G_7							0	$3G_4$
G_8								0

The entries below the main diagonal have been omitted because of the skewness of the Lie bracket.
i.e. G_1, G_2, G_3 form a closed subgroup. For I_2 and I_3 the same result is found. (This gives 9 in all, but one is a linear combination of the others.) Can use this subalgebra to find a linearising transformation.
From the commutation table for the generators of

$$y'' + 3yy' + y^3 = 0$$

we see that

$$[G_1, G_2] = -G_2, \qquad [G_1, G_3] = -G_3,$$
$$[G_2, G_3] = 0.$$

Define

$$[\widetilde{G}_1, \widetilde{G}_2] = [G_2, G_3] = 0,$$
$$[\widetilde{G}_1, \widetilde{G}_3] = [G_2, G_1] = G_2 = \widetilde{G}_1,$$
$$[\widetilde{G}_2, \widetilde{G}_3] = [G_3, G_1] = G_3 = \widetilde{G}_2.$$

i.e. we have the correct subalgebra.
In x, y coordinates we have

$$y'' + 3yy' + y^3 = 0,$$

$$\widetilde{G}_1 = y\frac{\partial}{\partial x} - y^3\frac{\partial}{\partial y},$$

$$\widetilde{G}_2 = xy\frac{\partial}{\partial x} - (xy^3 - y^2)\frac{\partial}{\partial y},$$

$$\widetilde{G}_3 = \frac{1}{2}x^2y\frac{\partial}{\partial x} - \left(\frac{1}{2}x^2y^3 - xy^2 + y\right)\frac{\partial}{\partial y}.$$

In X, Y coordinates we have

$$Y'' = 0,$$

$$G_1 = \frac{\partial}{\partial X},$$

$$G_2 = \frac{\partial}{\partial Y},$$

$$G_3 = X\frac{\partial}{\partial X} + Y\frac{\partial}{\partial Y}.$$

$$X = F(x,y), \qquad Y = G(x,y).$$

$$\frac{\partial}{\partial x} = \frac{\partial F}{\partial x}\frac{\partial}{\partial X} + \frac{\partial G}{\partial x}\frac{\partial}{\partial Y},$$

$$\frac{\partial}{\partial y} = \frac{\partial F}{\partial y}\frac{\partial}{\partial X} + \frac{\partial G}{\partial y}\frac{\partial}{\partial Y}.$$

1.

$$\left(y\frac{\partial F}{\partial x} - y^3\frac{\partial F}{\partial y}\right)\frac{\partial}{\partial X} + \left(y\frac{\partial G}{\partial x} - y^3\frac{\partial G}{\partial y}\right)\frac{\partial}{\partial Y} = \frac{\partial}{\partial X}.$$

$$y\frac{\partial G}{\partial x} - y^3\frac{\partial G}{\partial y} = 0.$$

$$\frac{dx}{y} = -\frac{dy}{y^3}.$$

$$0 = dx + \frac{dy}{y^2}; \qquad u = x - \frac{1}{y}.$$

$$G = G\left(x - \frac{1}{y}\right) = g\left(x - \frac{1}{y}\right).$$

$$y\frac{\partial F}{\partial x} - y^3\frac{\partial F}{\partial y} = 1.$$

$$\frac{dx}{y} = \frac{dy}{-y^3} = \frac{dF}{1}.$$

$$u = x - \frac{1}{y}.$$

(2^{nd} and 3^{rd}):

$$v = F - \frac{1}{2y^2}.$$

$$F = \frac{1}{2y^2} + f\left(x - \frac{1}{y}\right).$$

2.

$$\left[xy\frac{\partial F}{\partial x} - (xy^3 - y^2)\frac{\partial F}{\partial y}\right]\frac{\partial}{\partial X} + \left[xy\frac{\partial G}{\partial x} - (xy^3 - y^2)\frac{\partial G}{\partial y}\right]\frac{\partial}{\partial Y} = \frac{\partial}{\partial Y}.$$

F:

$$xyf' - (xy^3 - y^2)f'\left(\frac{1}{y^2}\right) = 0.$$

$$xyf' - xyf' + f' = 0.$$

$$f' = 0,$$

i.e. $f =$ constant (take as zero).

$$xyg' - (xy^3 - y^2)g'\left(\frac{1}{y^2}\right) = 1.$$

$$g' = 1.$$

$$g = x - \frac{1}{y},$$

to within a forgettable constant.

$$X = \frac{1}{2y^2}, \qquad Y = x - \frac{1}{y}.$$

$$y^2 = \frac{1}{2X}.$$

$$x = Y + (2X)^{1/2}.$$

$$y = \left(\frac{1}{2X}\right)^{1/2}.$$

$$y' = \frac{-\dfrac{\sqrt{2}}{4X^{3/2}}}{Y' + \dfrac{\sqrt{2}}{2X^{1/2}}}.$$

$$y'' = \frac{1}{\left(Y' + \dfrac{\sqrt{2}}{2X^{1/2}}\right)^3}\left\{\frac{\sqrt{2}}{4X^{3/2}}Y'' + \frac{3\sqrt{2}}{8X^{5/2}}Y' + \frac{1}{4X^3}\right\}.$$

$$y'' + 3yy' + y^3 = 0$$

$$\begin{aligned}
&Y'' + \frac{3}{2X}Y' + \frac{\sqrt{2}}{2X^{3/2}} \\
&\quad - \frac{3\sqrt{2}}{2X^{1/2}}\left(Y' + \frac{\sqrt{2}}{2X^{1/2}}\right)^2 + \left(Y' + \frac{\sqrt{2}}{2X^{1/2}}\right)^3 = 0.
\end{aligned}$$

$$\begin{aligned}
&Y'' + (Y')^3 + (Y')^2(0) + Y'\left(\frac{3}{2X} - \frac{3}{X} + \frac{3}{2X}\right) \\
&\qquad + \left(\frac{\sqrt{2}}{2X^{3/2}} - \frac{6\sqrt{2}}{8X^{3/2}} + \frac{2\sqrt{2}}{8X^{3/2}}\right) = 0.
\end{aligned}$$

Therefore

$$Y'' + (Y')^3 = 0.$$

This is not linear in the present ordering of the coordinates. The generators are

$$G_1 = \frac{\partial}{\partial X},$$
$$G_2 = \frac{\partial}{\partial Y},$$
$$G_3 = X\frac{\partial}{\partial X} + Y\frac{\partial}{\partial Y}.$$

Interchange the dependent and independent variables:

$$\mathcal{X} = Y, \qquad \mathcal{Y} = X.$$

Then

$$\mathcal{Y}' = \frac{1}{Y'}, \qquad \mathcal{Y}'' = -\frac{Y''}{(Y')^3}.$$

Hence

$$-\frac{\mathcal{Y}''}{(\mathcal{Y}')^3} + \frac{1}{(\mathcal{Y}')^3} = 0,$$

so

$$\mathcal{Y}'' = 1,$$

which is linear. To reduce it to the free-particle equation, put

$$W = \mathcal{Y} - \frac{1}{2}\mathcal{X}^2.$$

Then

$$W'' = 0.$$

Thus the corrected linearising transformation is

$$\mathcal{X} = x - \frac{1}{y}, \qquad W = \frac{1}{2y^2} - \frac{1}{2}\left(x - \frac{1}{y}\right)^2 = \frac{x}{y} - \frac{x^2}{2}.$$

$$Y(X) = A + BX, \qquad X = A + BY.$$

$$\frac{1}{2y^2} = A + B\left(x - \frac{1}{y}\right),$$

$$(A + Bx)y^2 - By - \frac{1}{2} = 0,$$

$$y = \frac{B \pm \sqrt{B^2 + 2(A + Bx)}}{2(A + Bx)}.$$

$$\widetilde{G}_1 = y\frac{\partial}{\partial x} - y^3\frac{\partial}{\partial y},$$

$$\widetilde{G}_2 = xy\frac{\partial}{\partial x} - (xy^3 - y^2)\frac{\partial}{\partial y}.$$

$$X = F(x,y), \qquad Y = G(x,y).$$

From

$$\widetilde{G}_1 \longrightarrow \frac{\partial}{\partial X},$$

$$F = \frac{1}{2y^2} + f\left(x - \frac{1}{y}\right), \qquad G = g\left(x - \frac{1}{y}\right).$$

$$\widetilde{G}_2 = \frac{\partial}{\partial Y}.$$

$$xy\frac{\partial F}{\partial x} - (xy^3 - y^2)\frac{\partial F}{\partial y} = 0.$$

$$xyf' - (xy^3 - y^2)\left(-\frac{1}{y^3} + \frac{1}{y^2}f'\right) = 0.$$

$$(xy - xy + 1)f' + x - \frac{1}{y} = 0.$$

$$f' = -\left(x - \frac{1}{y}\right),$$

$$f = -\frac{1}{2}\left(x - \frac{1}{y}\right)^2.$$

$$\begin{aligned} F &= \frac{1}{2y^2} - \frac{1}{2}\left(x - \frac{1}{y}\right)^2 \\ &= -\frac{1}{2}x^2 + \frac{x}{y}. \end{aligned}$$

$$xy\frac{\partial G}{\partial x} - (xy^3 - y^2)\frac{\partial G}{\partial y} = 1.$$

$$xyg' - (xy^3 - y^2)\frac{1}{y^2}g' = 1.$$

$$g' = 1, \qquad g = x - \frac{1}{y}.$$

Transform is

$$X = -\frac{1}{2}x^2 + \frac{x}{y}, \qquad Y = x - \frac{1}{y}.$$

$$Y' = \frac{1 + \dfrac{y'}{y^2}}{-x + \dfrac{1}{y} - \dfrac{xy'}{y^2}} = \frac{1}{\dfrac{y}{y^2 + y'} - x}.$$

$$\begin{aligned}
Y'' &= \frac{1}{-x + \dfrac{1}{y} - \dfrac{xy'}{y^2}} \left\{ -\frac{1}{\left(\dfrac{y}{y^2 + y'} - x\right)^2} \left[\frac{y'}{y^2 + y'} - \frac{y(2yy' + y'')}{(y^2 + y')^2} - 1 \right] \right\} \\
&= \frac{1}{(\cdots)} \left\{ y^2 y' + (y')^2 - 2y^2 y' - yy'' - y^4 - 2y^2 y' - (y')^2 \right\} \\
&= \frac{1}{(\cdots)} \left\{ -y \left(y'' + 3yy' + y^3 \right) \right\} = 0.
\end{aligned}$$

$$Y = A + BX.$$

$$x - \frac{1}{y} = A + B\left(-\frac{1}{2}x^2 + \frac{x}{y}\right).$$

$$(1 + Bx)\frac{1}{y} = \frac{1}{2}Bx^2 + x - A.$$

$$\begin{aligned}
y &= \frac{1 + Bx}{-A + x + \frac{1}{2}Bx^2} \\
&= \frac{2(I_1 + x)}{2I_2 + 2I_1 x + x^2},
\end{aligned}$$

$$I_1 = \frac{1}{B}, \qquad I_2 = -\frac{A}{B}.$$

Aside: differentiating y and solving for I_1 and I_2 between the two equations gives

$$I_1 = -x + \frac{y}{y' + y^2} \qquad \left(= \frac{1}{Y'}\right),$$

and

$$I_2 = \frac{1}{2}x^2 + \frac{1 - xy}{y' + y^2},$$

which are two resonance integrals.

$$y'' + 3yy' + y^3 = 0.$$

Is this the only equation of this structure which can be linearised?
Let

$$x = \alpha X, \qquad y = \beta Y.$$

Then

$$\frac{\beta}{\alpha^2}Y'' + 3\frac{\beta^2}{\alpha}YY' + \beta^3 Y^3 = 0,$$

or

$$Y'' + 3\alpha\beta YY' + (\alpha\beta)^2 Y^3 = 0.$$

Any choice of $\alpha\beta$ will give a different equation which can also be linearised.
Let

$$\begin{aligned} y &= \frac{\omega'}{\omega}, \\ y' &= \frac{\omega''\omega - (\omega')^2}{\omega^2}, \\ y'' &= \frac{\omega'''\omega^2 - 3\omega''\omega'\omega + 2(\omega')^3}{\omega^3}. \end{aligned}$$

Thus

$$y'' + 3yy' + y^3 \longrightarrow \frac{\omega'''}{\omega} = 0,$$

so

$$\omega''' = 0, \qquad \omega = A + Bx + Cx^2,$$

and

$$y = \frac{B + 2Cx}{A + Bx + Cx^2}.$$

This is a particular instance of a Riccati hierarchy. Recall the Riccati equation which is obtained from the general second-order linear differential equation

$$y'' + f(x)y' + g(x)y = 0.$$

Let

$$\frac{y'}{y} = u.$$

Then

$$\frac{y''}{y} - \frac{(y')^2}{y^2} = u', \qquad \frac{y''}{y} = u' + u^2,$$

and hence

$$u' + u^2 + f(x)u + g(x) = 0,$$

the Riccati equation.

What nonlinear differential equations can have $SL(3,\mathbb{R})$ symmetry?

It must be possible to transform the differential equation to

$$Q'' = 0$$

by means of a point transformation. The reverse also applies.
Let

$$T = F(q,t), \qquad Q = G(q,t).$$

Then $Q'' = 0$ becomes

$$\begin{aligned}
&\ddot{q} + \dot{q}^3 J^{-1}\left(\frac{\partial F}{\partial q}\frac{\partial^2 G}{\partial q^2} - \frac{\partial^2 F}{\partial q^2}\frac{\partial G}{\partial q}\right)\\
&\quad + \dot{q}^2 J^{-1}\left(\frac{\partial F}{\partial t}\frac{\partial^2 G}{\partial q^2} + 2\frac{\partial F}{\partial q}\frac{\partial^2 G}{\partial q\partial t} - 2\frac{\partial^2 F}{\partial q\partial t}\frac{\partial G}{\partial q} - \frac{\partial^2 F}{\partial q^2}\frac{\partial G}{\partial t}\right)\\
&\quad + \dot{q} J^{-1}\left(2\frac{\partial F}{\partial t}\frac{\partial^2 G}{\partial q\partial t} + \frac{\partial F}{\partial q}\frac{\partial^2 G}{\partial t^2} - \frac{\partial^2 F}{\partial t^2}\frac{\partial G}{\partial q} - 2\frac{\partial^2 F}{\partial q\partial t}\frac{\partial G}{\partial t}\right)\\
&\quad + J^{-1}\left(\frac{\partial F}{\partial t}\frac{\partial^2 G}{\partial t^2} - \frac{\partial^2 F}{\partial t^2}\frac{\partial G}{\partial t}\right) = 0,
\end{aligned}$$

where

$$J = \frac{\partial(F,G)}{\partial(t,q)} \neq 0.$$

The general third-order constant-coefficient equation

$$\omega''' + a\omega'' + b\omega' + c\omega = 0$$

has a solution. Let

$$\frac{\omega'}{\omega} = y.$$

Then

$$\frac{\omega'}{\omega} = y, \qquad \frac{\omega''}{\omega} = y' + y^2,$$

and

$$\begin{aligned}
\frac{\omega'''}{\omega} &= y'' + 2yy' + y(y' + y^2)\\
&= y'' + 3yy' + y^3.
\end{aligned}$$

Thus

$$y'' + 3yy' + y^3 + a(y' + y^2) + by + c = 0$$

can be linearised.
Aside: $\omega''' = 0$ has seven symmetries.

What second-order ODEs can be linearised?

$$y'' = f(x, y, y').$$

(Transparency.)

Classify by extra information

1. Existence of one symmetry Transform the differential equation to standard form by making the symmetry, say, $\partial/\partial x$. Then the differential equation is

$$y'' = f(y, y').$$

2. Existence of two symmetries The possibilities include

$$G_1 = x\frac{\partial}{\partial x}, \qquad G_2 = \frac{\partial}{\partial x}, \qquad [G_1, G_2] = -G_2,$$

and

$$G_1 = \frac{\partial}{\partial x}, \qquad G_2 = \frac{\partial}{\partial y}, \qquad [G_1, G_2] = 0,$$

with $G_2 \neq f(x, y)G_1$.

Two commuting symmetries

$$[G_1, G_2] = 0.$$

Suppose we choose coordinates such that

$$G_1 = \frac{\partial}{\partial y}, \qquad G_2 = \xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y}.$$

Then

$$[G_1, G_2] = \frac{\partial \xi}{\partial y}\frac{\partial}{\partial x} + \frac{\partial \eta}{\partial y}\frac{\partial}{\partial y} = 0,$$

so

$$\frac{\partial \xi}{\partial y} = 0, \qquad \frac{\partial \eta}{\partial y} = 0.$$

Observe that Type II is already linear and the solution is reduced to two quadratures.

Two non-commuting symmetries

$$[G_1, G_2] = G_1.$$

Let the coordinates be such that

$$G_1 = \frac{\partial}{\partial y}, \qquad G_2 = \xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y}.$$

Then $[G_1, G_2] = G_1$ becomes

$$\frac{\partial \xi}{\partial y}\frac{\partial}{\partial x} + \frac{\partial \eta}{\partial y}\frac{\partial}{\partial y} = \frac{\partial}{\partial y},$$

so

$$\frac{\partial \xi}{\partial y} = 0, \qquad \frac{\partial \eta}{\partial y} = 1.$$

Thus

$$\xi = a(x), \qquad \eta = b(x) + y,$$

and

$$G_2 = a(x)\frac{\partial}{\partial x} + (b(x) + y)\frac{\partial}{\partial y}.$$

The two forms written for G_2 are

$$\begin{aligned} G_2 &= y\frac{\partial}{\partial y} = yG_1, \\ G_2 &= x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} \neq f(x,y)G_1. \end{aligned}$$

Note that Type IV is linear. Type I,

$$y'' = f(y'),$$

and Type III,

$$y'' = \frac{1}{x}f(y'),$$

are the ones whose linearisation is still an open question.

Proposition

In order that a second-order ODE have the symmetry algebra $\mathfrak{sl}(3,\mathbb{R})$, and hence be linearisable, it is necessary and sufficient that it have the three-element subalgebra $\mathcal{X}$ of generators G_1, G_2, G_3 satisfying

$$[G_1, G_2] = 0, \qquad [G_2, G_3] = 0, \qquad [G_3, G_1] = G_2.$$

Proof

For

$$y'' = f(x, y, y'),$$

a generator of symmetry is

$$G = \xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y}.$$

We have three such generators, G_1, G_2, G_3. We cannot have both

$$G_1 = \phi(x,y)G_2, \qquad G_3 = \psi(x,y)G_2,$$

for suitable functions ϕ and ψ.
Suppose they are. Then

$$\begin{aligned}
[G_1, G_2] &= [\phi G_2, G_2] \\
&= \phi G_2 G_2 - G_2(\phi G_2) \\
&= \phi G_2 G_2 - \phi G_2 G_2 - (G_2 \phi) G_2 \\
&= 0.
\end{aligned}$$

if $G_2\phi = 0$. Likewise,

$$[G_2, G_3] = 0 \quad \Longrightarrow \quad G_2\psi = 0.$$

Then

$$\begin{aligned}
[G_3, G_1] &= [\psi G_2, \phi G_2] \\
&= \psi G_2(\phi G_2) - \phi G_2(\psi G_2) \\
&= \psi\phi G_2 G_2 + \psi(G_2\phi)G_2 - \phi\psi G_2 G_2 - \phi(G_2\psi)G_2 \\
&= 0 \neq G_2.
\end{aligned}$$

We assume that there is no function $\phi(x,y)$ such that

$$G_1 = \phi G_2.$$

(We cannot make the statement about both G_1 and G_3.) Hence there exists a regular point transformation

$$Y = F(x,y), \qquad X = G(x,y),$$

such that the generators are

$$G_1 = \frac{\partial}{\partial x}, \qquad G_2 = \frac{\partial}{\partial y}.$$

Write

$$G_3 = \xi\frac{\partial}{\partial x} + \eta\frac{\partial}{\partial y}.$$

Then

$$\begin{aligned}
[G_2, G_3] &= \frac{\partial \xi}{\partial y}\frac{\partial}{\partial x} + \frac{\partial \eta}{\partial y}\frac{\partial}{\partial y} &&= 0, \\
[G_3, G_1] &= -\frac{\partial \xi}{\partial x}\frac{\partial}{\partial x} - \frac{\partial \eta}{\partial x}\frac{\partial}{\partial y} &&= \frac{\partial}{\partial y}.
\end{aligned}$$

That is,

$$\frac{\partial \xi}{\partial x} = 0, \qquad \frac{\partial \xi}{\partial y} = 0, \qquad \frac{\partial \eta}{\partial x} = -1, \qquad \frac{\partial \eta}{\partial y} = 0.$$

Thus

$$G_3 = A\frac{\partial}{\partial x} + (B - x)\frac{\partial}{\partial y}.$$

Take

$$G_3 = -x\frac{\partial}{\partial y}.$$

The equation

$$y'' = f(x, y, y')$$

is invariant under these three generators. From

$$G_1^{(2)}[y'' - f(x, y, y')] = 0$$

we obtain

$$\frac{\partial f}{\partial x} = 0.$$

From

$$G_2^{(2)}[y'' - f] = 0$$

we obtain

$$\frac{\partial f}{\partial y} = 0.$$

Finally,

$$G_3^{(2)}[y'' - f] = 0$$

gives

$$\left(-x\frac{\partial}{\partial y} - \frac{\partial}{\partial y'}\right)[y'' - f] = 0,$$

so

$$x\frac{\partial f}{\partial y} + \frac{\partial f}{\partial y'} = 0.$$

Therefore f is constant and the differential equation is

$$y'' = \text{constant},$$

which is linear. Hence the symmetry of the original differential equation is $\mathfrak{sl}(3, \mathbb{R})$. We have proven sufficiency, i.e. the possession of G_1, G_2, G_3 such that

$$[G_1, G_2] = 0, \qquad [G_2, G_3] = 0, \qquad [G_3, G_1] = G_2.$$

Necessity follows from $\mathcal{X}$ being a subalgebra of $\mathfrak{sl}(3, \mathbb{R})$.

Can one do better?

Suppose we have two generators G_1 and G_2 with the properties

$$[G_1, G_2] = 0, \qquad G_2 = f(x,y)G_1.$$

Make a point transformation to X, Y so that

$$\overline{G}_1 = \frac{\partial}{\partial Y}, \qquad G_2 = F(X,Y)\frac{\partial}{\partial Y}.$$

Then

$$[G_1, G_2] = \left[\frac{\partial}{\partial Y}, F\frac{\partial}{\partial Y}\right] = \frac{\partial F}{\partial Y}\frac{\partial}{\partial Y} = 0,$$

and hence

$$\frac{\partial F}{\partial Y} = 0.$$

Without loss of generality, take $F = X$. Thus

$$G_1 = \frac{\partial}{\partial y}, \qquad G_2 = x\frac{\partial}{\partial y},$$

which means that $y'' = F(x, y, y')$ has the form

$$y'' = F(x),$$

which is linear and so has $\mathfrak{sl}(3, \mathbb{R})$ symmetry.
If we take G_1 and G_2 such that

$$[G_1, G_2] = 0, \qquad G_2 \neq f(x,y)G_1,$$

we may take the standard forms

$$G_1 = \frac{\partial}{\partial x}, \qquad G_2 = \frac{\partial}{\partial y}.$$

The differential equation invariant under G_1 and G_2 is

$$y'' = f(y').$$

We know that linearisation is possible only if the differential equation has the form

$$y'' + ()y'^3 + ()y'^2 + ()y' + () = 0,$$

where the quantities in parentheses are functions of x and y. Hence $f(y')$ must take the form

$$f(y') = a(y')^3 + b(y')^2 + cy' + d,$$

where a, b, c, d are constants. Thus

$$y'' = a(y')^3 + b(y')^2 + cy' + d, \qquad a \neq 0.$$

Put

$$y = \alpha Y + \beta X, \qquad x = \gamma X.$$

Then

$$\frac{\alpha}{\gamma^2} Y'' = a\left(\frac{\alpha}{\gamma} Y' + \beta\right)^3 + b\left(\frac{\alpha}{\gamma} Y' + \beta\right)^2 + c\left(\frac{\alpha}{\gamma} Y' + \beta\right) + d.$$

The choice

$$\beta = -\frac{b}{3a}$$

removes the $(Y')^2$ term, and

$$\frac{a\alpha^2}{\gamma} = 1$$

makes the coefficient of $(Y')^3$ unity. The equation reduces to

$$y'' = (y')^3 + cy' + d.$$

For this to be linearisable there must be six more generators in addition to the two we already have. Let

$$G = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}$$

be one of them. Then

$$\begin{aligned}
&\eta_{xx} + 2y'\eta_{xy} + (y')^2\eta_{yy} + \Big((y')^3 + cy' + d\Big)\eta_y \\
&\quad - y'\left[\xi_{xx} + 2y'\xi_{xy} + (y')^2\xi_{yy} + \Big((y')^3 + cy' + d\Big)\xi_y\right] \\
&\quad - 2\Big((y')^3 + cy' + d\Big)(\xi_x + y'\xi_y) \\
&= (3(y')^2 + c)\left(\eta_x + y'\eta_y - y'\xi_x - (y')^2\xi_y\right).
\end{aligned}$$

Equating coefficients of like powers of y' gives

$$\xi_{yy} + 2\eta_y - \xi_x = 0, \tag{i}$$

$$2c\xi_y + 3\eta_x - \eta_{yy} + 2\xi_{xy} = 0, \tag{ii}$$

$$c\xi_x + 3d\xi_y - 2\eta_{xy} + \xi_{xx} = 0, \tag{iii}$$

$$c\eta_x - d\eta_y + 2d\xi_x - \eta_{xx} = 0. \tag{iv}$$

From (iv),

$$\xi_x = \frac{1}{2d}\left(\eta_{xx} + d\eta_y - c\eta_x\right),$$

and from (iii),

$$\begin{aligned}\xi_y &= \frac{1}{3d}\left(2\eta_{xy} - \xi_{xx} - c\xi_x\right)\\ &= \frac{1}{3d}\left[2\eta_{xy} - \frac{1}{2d}\left(\eta_{xxx} + d\eta_{xy} - c\eta_{xx}\right) - \frac{c}{2d}\left(\eta_{xx} + d\eta_y - c\eta_x\right)\right].\end{aligned}$$

Eliminate η. From (i),

$$\eta_y = \frac{1}{2}(\xi_x - \xi_{yy}), \tag{v}$$

and from (ii),

$$\begin{aligned}\eta_x &= \frac{1}{3}(\eta_{yy} - 2c\xi_y - 2\xi_{xy})\\ &= -\frac{1}{6}(\xi_{yyy} + 3\xi_{xy} + 4c\xi_y),\end{aligned} \tag{vi}$$

using (v). Substitution of (v) and (vi) in (iii) and (iv) gives

$$\xi_{xyy} + 3d\xi_y + c\xi_x = 0, \tag{vii}$$

and

$$\xi_{xyyy} - c\xi_{yyy} + 3\xi_{xxy} + 3d\xi_{yy} + c\xi_{xy} - 4c^2\xi_y + 9d\xi_x = 0. \tag{viii}$$

The consistency condition between (v) and (vi) is

$$\frac{\partial}{\partial x}(\eta_y) = \frac{\partial}{\partial y}(\eta_x),$$

which gives

$$\xi_{yyyy} + 4c\xi_{yy} + 3\xi_{xx} = 0. \tag{ix}$$

Using the y derivative of (vii), equation (viii) simplifies to

$$-c\xi_{yyy} + 3\xi_{xxy} - 4c^2\xi_y + 9d\xi_x = 0. \tag{x}$$

Also,

$$\frac{\partial^2}{\partial y^2}(\text{vii}) - \frac{\partial}{\partial x}(\text{ix})$$

gives

$$\xi_{xxx} + c\xi_{xyy} - d\xi_{yyy} = 0. \tag{xi}$$

Use (vii), (x), and (xi), which can be shown to be equivalent to (vii), (viii), and (ix), except when both c and d are zero.

For the case $c = d = 0$,

$$\xi_{xyy} = 0, \tag{vii}$$

and

$$\xi_{xyyy} + 3\xi_{xxy} = 0. \tag{viii}$$

Equation (ix) becomes

$$\xi_{yyyy} + 3\xi_{xx} = 0. \tag{ix}$$

From (vii),

$$\xi = F(y) + yG(x) + H(x).$$

Equation (viii) gives

$$G'' = 0.$$

In (ix),

$$F^{(4)} + 3yG'' + H'' = 0.$$

Therefore

$$\begin{aligned} G &= G_0 + G_1 x, \\ H &= H_0 + H_1 x + \frac{H_2}{2}x^2, \\ F &= F_1 + F_2 y + F_3 y^2 + F_4 y^3 - \frac{H_2}{24}y^4. \end{aligned}$$

Thus

$$\begin{aligned} \xi &= F_1 + F_2 y + F_3 y^2 + F_4 y^3 - \frac{H_2}{24}y^4 + G_0 y + G_1 xy \\ &\quad + H_0 + H_1 x + \frac{H_2}{2}x^2 \\ &= (F_1 + H_0) + (F_2 + G_0)y + F_3 y^2 + F_4 y^3 + G_1 xy + H_1 x \\ &\quad + H_2\left(\frac{1}{2}x^2 - \frac{1}{24}y^4\right). \end{aligned}$$

The eighth generator comes from

$$\eta_x = 0, \qquad \eta_y = 0$$

when $\xi = 0$, i.e. η is constant. This gives

$$G_1 = \frac{\partial}{\partial y}.$$

For

$$y'' = (y')^3,$$

find the linearising transformation.
Note that we have

$$\frac{\partial}{\partial x}, \qquad \frac{\partial}{\partial y}, \qquad -y\frac{\partial}{\partial x}.$$

Let $X = y$ and $Y = x$. Then the algebra consists of

$$\frac{\partial}{\partial Y}, \qquad \frac{\partial}{\partial X}, \qquad -X\frac{\partial}{\partial Y},$$

which is the standard form. Also,

$$Y' = \frac{1}{y'}, \qquad Y'' = -\frac{y''}{(y')^3}.$$

Thus $y'' = (y')^3$ becomes

$$-\frac{Y''}{(Y')^3} = \frac{1}{(Y')^3},$$

or

$$Y'' = -1.$$

Hence

$$\begin{aligned} Y &= A + BX - \frac{1}{2}X^2, \\ x &= A + By - \frac{1}{2}y^2. \end{aligned}$$

Alternatively,

$$\begin{aligned} y'' &= (y')^3, \\ \frac{y''}{(y')^2} &= y', \\ -\frac{1}{y'} &= y + C, \\ -1 &= (y + C)y'. \end{aligned}$$

and

$$K - x = \frac{1}{2}y^2 + Cy.$$

Retain

$$\xi_{xxx} + c\xi_{xyy} - d\xi_{yyy} = 0, \tag{xi}$$

$$\xi_{xyy} + 3d\xi_y + c\xi_x = 0, \tag{vii}$$

and

$$-c\xi_{yyy} + 3\xi_{xxy} - 4c^2\xi_y + 9d\xi_x = 0. \tag{x}$$

Observe that (xi) is homogeneous in order. Hence it can be solved by factorisation. Suppose it factors as

$$\left(\frac{\partial}{\partial x}+\alpha\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x}+\beta\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x}+\gamma\frac{\partial}{\partial y}\right)\xi=0.$$

Corresponding to each factor there is an equation for a characteristic. However, one must be careful of repeated roots. Observe that

$$\alpha\beta\gamma=-d,\qquad \alpha\beta+\beta\gamma+\gamma\alpha=c,\qquad \alpha+\beta+\gamma=0.$$

A triple root gives $c=d=0$, which has already been treated.

For two roots α and -2α,

$$c=-3\alpha^2,\qquad d=-2\alpha^3,$$

and

$$\left(\frac{\partial}{\partial x}+\alpha\frac{\partial}{\partial y}\right)^2\left(\frac{\partial}{\partial x}-2\alpha\frac{\partial}{\partial y}\right)\xi=0.$$

The characteristic equations give

$$u_1=\alpha x-y,\qquad u_2=2\alpha x+y.$$

The solution is

$$\xi=f(\alpha x-y)+g(2\alpha x+y)+(2\alpha x+y)h(\alpha x-y).$$

Substitution in (xi) and (x) gives

$$\begin{aligned}&\alpha f'''+2\alpha g'''+2\alpha h''+\alpha h''-\alpha h'''+\alpha(2\alpha x+y)h'''\\&\quad-6\alpha^3[-f'+g'+h-(2\alpha x+y)h']\\&\quad-3\alpha^2[\alpha f''+2\alpha g''+2\alpha h+\alpha(2\alpha x+y)h'']=0.\end{aligned}$$

The separate coefficients give

$$2\alpha g'''-12\alpha^3g'=0,$$

$$(2\alpha x+y)(h'''+3\alpha^3h')=0,$$

and

$$\alpha f'''+2\alpha h''+3\alpha^3f'=0.$$

The remaining determining equation is

$$\begin{aligned}0&=3\alpha^2\left[-f'''+g'''+3h''-(2\alpha x+y)h'''\right]\\&\quad+3\left[-\alpha^2f'''+4\alpha^2g'''-3\alpha^2h''-\alpha^2(2\alpha x+y)h'''\right]\\&\quad+36\alpha^4\left[-f'+g'+h-(2\alpha x+y)h'\right]\\&\quad-18\alpha^3\left[\alpha f''+2\alpha g''+2\alpha h'+\alpha(2\alpha x+y)h''\right].\end{aligned}\tag{ix}$$

The correct equations are

$$g''' = 0, \qquad h''' + 9\alpha^2 h' = 0, \qquad f''' + 9\alpha^2 f' = 0.$$

The solutions written in the notes are

$$\begin{aligned}
g &= G_0 + G_1(2\alpha x + y) + G_2(2\alpha x + y)^2,\\
h &= [H_0 + H_1 \sin 3\alpha(\alpha x - y) + H_2 \cos 3\alpha(\alpha x - y)]\,(2\alpha x + y),\\
f &= f_0 + F_1(\alpha x - y) + F_2(\alpha x - y)^2.
\end{aligned}$$

There are seven independent solutions and so seven generators. The eighth comes from $\xi = 0$, giving η constant.

For three distinct roots,

$$-2\alpha, \qquad \alpha + \delta, \qquad \alpha - \delta,$$

where $\delta \neq 0, \pm 3\alpha$ to avoid repeated roots, equation (xi) becomes

$$\left(\frac{\partial}{\partial x} - 2\alpha \frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} + (\alpha + \delta)\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} + (\alpha - \delta)\frac{\partial}{\partial y}\right)\xi = 0,$$

where

$$c = \delta^2 - 3\alpha^2, \qquad d = 2\alpha(\alpha^2 - \delta^2).$$

The characteristics give

$$u_1 = 2\alpha x + y, \qquad u_2 = (\alpha + \delta)x - y, \qquad u_3 = (\alpha - \delta)x - y,$$

and

$$\xi = f(2\alpha x + y) + g((\alpha + \delta)x - y) + h((\alpha - \delta)x - y).$$

Substitution into (vii) and (xi) again gives seven generators, which become eight with η constant. Hence

$$y'' = (y')^3 + cy' + d$$

is always linearisable.

From Type 2, consider

$$y'' = (y')^2 + d.$$

Invariance under

$$G = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y}$$

gives

$$
\begin{aligned}
&\eta_{xx} + 2y'\eta_{xy} + (y')^2\eta_{yy} + \Big((y')^2 + d\Big)\eta_y \\
&\quad - y'\left[\xi_{xx} + 2y'\xi_{xy} + (y')^2\xi_{yy} + \Big((y')^2 + d\Big)\xi_y\right] \\
&\quad - 2\Big((y')^2 + d\Big)(\xi_x + y'\xi_y) \\
&= 2y'\left(\eta_x + y'\eta_y - y'\xi_x - (y')^2\xi_y\right).
\end{aligned}
$$

Equating powers of y' gives

$$\xi_{yy} + \xi_y = 0, \tag{i}$$

$$\eta_{yy} - \eta_y - 2\xi_{xy} = 0, \tag{ii}$$

$$2\eta_{xy} - 2\eta_x - \xi_{xx} - 3d\xi_y = 0, \tag{iii}$$

$$\eta_{xx} + d\eta_y - 2d\xi_x = 0. \tag{iv}$$

From (i),

$$\xi = a(x) + b(x)e^{-y}.$$

In (ii),

$$\eta_{yy} - \eta_y = -2b'e^{-y},$$

so

$$\eta = c(x) + d(x)e^{y} - b'(x)e^{-y}.$$

Substitution in (iii) gives

$$2[d'e^y + b''e^{-y}] - 2[c' + d'e^y - b''e^{-y}] - [a'' + b''e^{-y}] + 3Dbe^{-y} = 0.$$

Thus

$$3(b'' + Db) = 0, \tag{v}$$

$$a'' + 2c' = 0. \tag{vi}$$

Substitution in (iv) gives

$$c'' + d''e^y - b'''e^{-y} + D[de^y + b'e^{-y}] - 2D[a' + b'e^{-y}] = 0,$$

so

$$b''' + Db' = 0 \qquad \text{(redundant)},$$

$$d'' + Dd = 0, \tag{vii}$$

and

$$c'' - 2Da' = 0. \tag{viii}$$

Equations (v), (vi), and (viii) give

$$a''' + 4Da' = 0,$$

and

$$c' = C_1 + 2Da.$$

There appear to be nine constants, so one must be lost. Write

$$\begin{aligned} a &= A_0 + A_1 f_1 + A_2 f_2, \\ c' &= (C_1 + A_0) + A_1 f_1 + A_2 f_2, \\ c &= C_0 + (C_1 + A_0)x + A_1 \int f_1\, dx + A_2 \int f_2\, dx. \end{aligned}$$

Thus all differential equations of the form

$$y'' = (y')^2 + d$$

have $\mathfrak{sl}(3, \mathbb{R})$ symmetry and are accordingly linearisable. Writing $D = \omega^2$, the generators are

$$\begin{aligned} G_1 &= \frac{\partial}{\partial y}, \\ G_2 &= \frac{\partial}{\partial x}, \\ G_3 &= e^{-2\omega x}\frac{\partial}{\partial x} + \omega e^{-2\omega x}\frac{\partial}{\partial y}, \\ G_4 &= e^{2\omega x}\frac{\partial}{\partial x} - \omega e^{2\omega x}\frac{\partial}{\partial y}, \\ G_5 &= e^{-y+\omega x}\frac{\partial}{\partial x} - \omega e^{-y+\omega x}\frac{\partial}{\partial y}, \\ G_6 &= e^{-y-\omega x}\frac{\partial}{\partial x} + \omega e^{-y-\omega x}\frac{\partial}{\partial y}, \\ G_7 &= e^{y-\omega x}\frac{\partial}{\partial y}, \\ G_8 &= e^{y+\omega x}\frac{\partial}{\partial y}. \end{aligned}$$

The set $\{G_3, G_6, G_7\}$ has the algebra $\mathcal{X}$:

$$[G_3, G_6] = 0, \qquad [G_6, G_7] = -G_3, \qquad [G_7, G_3] = 0.$$

The standard form of $\mathcal{X}$ is as on page 12:

$$[\widetilde{G}_1, \widetilde{G}_2] = 0, \qquad [\widetilde{G}_2, \widetilde{G}_3] = 0, \qquad [\widetilde{G}_3, \widetilde{G}_1] = \widetilde{G}_2,$$

with

$$\widetilde{G}_1 = \frac{\partial}{\partial x}, \qquad \widetilde{G}_2 = \frac{\partial}{\partial y}, \qquad \widetilde{G}_3 = -x\frac{\partial}{\partial y}.$$

Try

$$G_6 \longrightarrow -\widetilde{G}_3, \qquad G_7 \longrightarrow \widetilde{G}_1, \qquad G_3 \longrightarrow \widetilde{G}_2.$$

Let

$$X = F(x,y), \qquad Y = G(x,y).$$

The mapping of G_7 gives

$$e^{y-\omega x}\left[\frac{\partial F}{\partial y}\frac{\partial}{\partial X} + \frac{\partial G}{\partial y}\frac{\partial}{\partial Y}\right] = \frac{\partial}{\partial X}.$$

Hence

$$\frac{\partial F}{\partial y} = e^{-y+\omega x}, \qquad F = k(x) - e^{-y+\omega x},$$

and

$$\frac{\partial G}{\partial y} = 0, \qquad G = m(x).$$

The mapping of G_6 gives

$$e^{-y-\omega x}\left(\frac{\partial F}{\partial x} + \omega\frac{\partial F}{\partial y}\right) = 0, \tag{*}$$

and

$$e^{-y-\omega x}\left(\frac{\partial G}{\partial x} + \omega\frac{\partial G}{\partial y}\right) = F. \tag{†}$$

Substitution in (*) gives

$$k' - \omega e^{-y+\omega x} + \omega e^{-y+\omega x} = 0,$$

so $k' = 0$. Equation (†) gives

$$e^{-y-\omega x} m' = k - e^{-y+\omega x},$$

or

$$m' = k e^{y+\omega x} - e^{2\omega x}.$$

Thus $k = 0$, and the notes take

$$F = -e^{-y+\omega x}, \qquad G = -e^{2\omega x}.$$

Check:

$$Y' = \frac{-2\omega e^{2\omega x}}{(y'-\omega)e^{-y+\omega x}},$$

$$Y'' = \frac{-4\omega^2 e^{2\omega x}(y'-\omega)e^{-y+\omega x} + 2\omega e^{2\omega x}\left[y'' - (y'-\omega)^2\right]e^{-y+\omega x}}{\left[(y'-\omega)e^{-y+\omega x}\right]^3}.$$

The numerator is

$$\begin{aligned} &2\omega e^{-y+3\omega x}\left[-2\omega y' + 2\omega^2 + y'' - (y')^2 + 2\omega y' - \omega^2\right] \\ &= 2\omega e^{-y+3\omega x}\left[y'' - (y')^2 + \omega^2\right], \end{aligned}$$

as required.
The solution of $Y'' = 0$ is

$$Y = A + BX.$$

The solution of $y'' = (y')^2 - \omega^2$ is therefore

$$\begin{aligned} -e^{2\omega x} &= A + Be^{-y+\omega x}, \\ e^{-y} &= Ce^{\omega x} + De^{-\omega x}, \\ y &= -\log\left(Ce^{\omega x} + De^{-\omega x}\right). \end{aligned}$$

Check:

$$\begin{aligned} \frac{y''}{(y')^2 - \omega^2} &= 1, \\ \frac{d\left((y')^2\right)}{(y')^2 - \omega^2} &= 2\,dy, \\ \log\left((y')^2 - \omega^2\right) &= 2y + k, \\ (y')^2 &= \omega^2 + me^{2y}. \end{aligned}$$

Thus

$$I = \int \frac{dy}{\left(\omega^2 + me^{2y}\right)^{1/2}} = \int dx.$$

Let

$$e^{-y} = u, \qquad -e^{-y}\,dy = du, \qquad dy = -\frac{du}{u}.$$

Then

$$I = \int \frac{-u^{-1}\,du}{\left(\omega^2 + mu^{-2}\right)^{1/2}} = -\int \frac{du}{\left(\omega^2 u^2 + m\right)^{1/2}}.$$

$$I = -\frac{1}{\omega}\operatorname{arsinh}\left(\frac{\omega u}{m^{1/2}}\right) = x + c.$$

Therefore

$$u = -\frac{m^{1/2}}{\omega}\sinh(\omega x + \alpha),$$
$$y = -\log\left[-\frac{m^{1/2}}{\omega}\sinh(\omega x + \alpha)\right].$$

As an aside,

$$\int \frac{dx}{(a^2 + e^{2x})^{1/2}} = \int \frac{e^{-x}\,dx}{(a^2 e^{-2x} + 1)^{1/2}} = \int \frac{-d(e^{-x})}{(a^2 e^{-2x} + 1)^{1/2}}.$$

Type III

$$xy'' = f(y'), \qquad f(y') = a(y')^3 + b(y')^2 + cy' + d.$$

There are two standard forms:

$$\begin{aligned} xy'' &= (y')^3 + cy' + d, && a \neq 0, \\ xy'' &= (y')^2 + d, && a = 0. \end{aligned}$$

Using the same type of analysis, it is found that only

$$xy'' = (y')^3 + y'$$

is linearisable.
Note that the equation

$$xy'' = (y')^3 - \frac{1}{2}y'$$

does have $\mathfrak{sl}(2, \mathbb{R})$ symmetry.

Linearisation of a system of equations

Theorem

A system of n equations of the form

$$A\mathbf{u}'' = \mathbf{f}(x, \mathbf{u}, \mathbf{u}'), \tag{1}$$

where A is a constant matrix, is linearisable if it possesses the algebra $\mathcal{N}$:

$$\begin{aligned} [H, G_i] &= 0, && (2) \\ [G_i, G_j] &= 0, && (3) \\ [G_i, X_j] &= 0, && (4) \\ [H, X_i] &= G_i. && (5) \end{aligned}$$

where $\{H, G_i, X_i \mid i = 1, \ldots, n\}$ are symmetries of (1).

Proof

We cannot have both

$$H = a_i G_i \qquad \text{and} \qquad X_i = b_{ij} G_j$$

for suitable functions a_i and b_{ij}. For, from (2),

$$\begin{aligned}[H, G_j] &= [a_i G_i, G_j] \\ &= a_i [G_i, G_j] - (G_j a_i) G_i \\ &= -(G_j a_i) G_i \\ &= 0,\end{aligned}$$

and, from (4),

$$\begin{aligned}[G_i, X_j] &= [G_i, b_{jk} G_k] \\ &= [G_i, G_k] b_{jk} + (G_i b_{jk}) G_k \\ &= (G_i b_{jk}) G_k \\ &= 0.\end{aligned}$$

Then (5) is

$$\begin{aligned}[H, X_j] &= [a_i G_i, b_{jk} G_k] \\ &= a_i [G_i, G_k] b_{jk} + a_i (G_i b_{jk}) G_k - b_{jk} (G_k a_i) G_i \\ &= 0,\end{aligned}$$

which is a contradiction.

We assume that $H \neq a_i G_i$. Then there exist coordinates x, u_i such that

$$H = \frac{\partial}{\partial x}, \qquad G_j = C_{ij} \frac{\partial}{\partial u_i}.$$

Since

$$[H, G_j] = \frac{\partial C_{ij}}{\partial x} \frac{\partial}{\partial u_i} = 0,$$

it follows that

$$\frac{\partial C_{ij}}{\partial x} = 0,$$

so $C_{ij} = C_{ij}(\mathbf{u})$. Since the G_i are linearly independent, the matrix $[C_{ij}]$ has rank n, and so its inverse exists, perhaps restricted to a subset of $\mathbb{R}^n$.

If we change variables from u_i to v_i so that

$$v_i = d_i(\mathbf{u}),$$

then

$$\begin{aligned} G_i &= C_{ik}\frac{\partial}{\partial u_k} \\ &= C_{ik}\frac{\partial d_j}{\partial u_k}\frac{\partial}{\partial v_j} \\ &= \frac{\partial}{\partial v_i}, \end{aligned}$$

provided

$$C_{ik}\frac{\partial d_j}{\partial u_k} = \delta_{ij},$$

i.e. the Jacobian of the transformation is the inverse of $[C_{ij}]$. Thus the transformation is not degenerate. Can functions $d_j(\mathbf{u})$ exist with the property

$$\frac{\partial d_i}{\partial u_j} = (C^{-1})_{ij}, \qquad C = [C_{ij}]?$$

The requirement of consistency, if the integration is to be carried out, is

$$\frac{\partial}{\partial u_k}\frac{\partial d_i}{\partial u_j} = \frac{\partial}{\partial u_j}\frac{\partial d_i}{\partial u_k}.$$

Now

$$\frac{\partial C^{-1}}{\partial u_i} = -C^{-1}\frac{\partial C}{\partial u_i}C^{-1}.$$

The differentiability of C^{-1} depends upon the differentiability of C. Since we need C'' in $G^{(2)}$, C must be twice differentiable almost everywhere, and so d_i is three times differentiable and the mixed derivatives are equal. Thus a transformation exists which casts G_i in the form

$$G_i = \frac{\partial}{\partial u_i}.$$

Had

$$X_i = b_{ij}\frac{\partial}{\partial u_j},$$

then

$$\begin{aligned} [G_i, X_j] &= \left[\frac{\partial}{\partial u_i}, b_{jk}\frac{\partial}{\partial u_k}\right] \\ &= \frac{\partial b_{jk}}{\partial u_i}\frac{\partial}{\partial u_k} = 0. \end{aligned}$$

Hence

$$\frac{\partial b_{jk}}{\partial u_i} = 0, \qquad b_{jk} = B_{jk}(x).$$

From (5),

$$
\begin{aligned}
[H, X_i] &= \left[\frac{\partial}{\partial x}, B_{ij}\frac{\partial}{\partial u_j}\right] \\
&= \frac{dB_{ij}}{dx}\frac{\partial}{\partial u_j} = \frac{\partial}{\partial u_i}.
\end{aligned}
$$

Therefore

$$
B'_{ij} = \delta_{ij}, \qquad B_{ij}(x) = \delta_{ij}x + K_{ij}.
$$

Thus

$$
X_i = x\frac{\partial}{\partial u_i} + K_{ij}\frac{\partial}{\partial u_j} = x\frac{\partial}{\partial u_i} + K_{ij}G_j.
$$

Drop the second term and take

$$
X_i = x\frac{\partial}{\partial u_i}.
$$

We have the standard form

$$
H = \frac{\partial}{\partial x}, \qquad G_i = \frac{\partial}{\partial u_i}, \qquad X_i = x\frac{\partial}{\partial u_i}.
$$

In the coordinates (x, u_i), the differential equation is

$$
A\mathbf{u}'' = \mathbf{f}(x, \mathbf{u}, \mathbf{u}').
$$

Invariance under H, G_i, and X_i implies, respectively,

$$
\frac{\partial \mathbf{f}}{\partial x} = 0, \qquad \frac{\partial \mathbf{f}}{\partial \mathbf{u}} = 0, \qquad \frac{\partial \mathbf{f}}{\partial \mathbf{u}'} = 0.
$$

For example, consider

$$
u'' + \frac{u(v')^2}{v^2} = 0,
$$

$$
v'' + \frac{2u'v'}{u} - \frac{(v')^2}{v} = 0.
$$

Let

$$
G = \tau(u, v, x)\frac{\partial}{\partial x} + \xi(u, v, x)\frac{\partial}{\partial u} + \eta(u, v, x)\frac{\partial}{\partial v}.
$$

Then

$$
\begin{aligned}
G^{(2)} = G &+ (\xi' - u'\tau')\frac{\partial}{\partial u'} + (\eta' - v'\tau')\frac{\partial}{\partial v'} \\
&+ (\xi'' - u'\tau'' - 2u''\tau')\frac{\partial}{\partial u''} + (\eta'' - v'\tau'' - 2v''\tau')\frac{\partial}{\partial v''}.
\end{aligned}
$$

For example,

$$\eta' = \frac{\partial \eta}{\partial x} + u'\frac{\partial \eta}{\partial u} + v'\frac{\partial \eta}{\partial v},$$

and

$$\begin{aligned}\eta'' = &\frac{\partial^2 \eta}{\partial x^2} + 2u'\frac{\partial^2 \eta}{\partial u \partial x} + 2v'\frac{\partial^2 \eta}{\partial v \partial x} + (u')^2\frac{\partial^2 \eta}{\partial u^2}\\ &+ 2u'v'\frac{\partial^2 \eta}{\partial u \partial v} + (v')^2\frac{\partial^2 \eta}{\partial v^2} + u''\frac{\partial \eta}{\partial u} + v''\frac{\partial \eta}{\partial v}.\end{aligned}$$

The action of $G^{(2)}$ on the two equations, followed by separation by powers of u' and v', leads to a system of fifteen independent partial differential equations for τ, ξ, and η, all linear. The solutions give

$$\begin{aligned}
G_1 &= x\frac{\partial}{\partial x} + 2u\frac{\partial}{\partial u} + 2v\frac{\partial}{\partial v},\\
G_2 &= \frac{\partial}{\partial x},\\
G_3 &= \frac{1}{v}\frac{\partial}{\partial u} + \frac{1}{u}\frac{\partial}{\partial v},\\
G_4 &= x\frac{\partial}{\partial x},\\
G_5 &= x^2\frac{\partial}{\partial x} + 2xu\frac{\partial}{\partial u} + 2xv\frac{\partial}{\partial v},\\
G_6 &= \frac{x}{v}\frac{\partial}{\partial u} + \frac{x}{u}\frac{\partial}{\partial v},\\
G_7 &= xuv\frac{\partial}{\partial x} + 2u^2v\frac{\partial}{\partial u} + 2uv^2\frac{\partial}{\partial v},\\
G_8 &= uv\frac{\partial}{\partial x},\\
G_9 &= x\frac{\partial}{\partial x} + 2u\frac{\partial}{\partial u} - 2v\frac{\partial}{\partial v},\\
G_{10} &= v\frac{\partial}{\partial u} - \frac{v^2}{u}\frac{\partial}{\partial v},\\
G_{11} &= x^2\frac{\partial}{\partial x} + 2ux\frac{\partial}{\partial u} - 2vx\frac{\partial}{\partial v},\\
G_{12} &= vx\frac{\partial}{\partial u} - \frac{v^2x}{u}\frac{\partial}{\partial v},\\
G_{13} &= \frac{xv}{u}\frac{\partial}{\partial x} + \frac{2u^2}{v}\frac{\partial}{\partial u} - 2u\frac{\partial}{\partial v},\\
G_{14} &= \frac{u}{v}\frac{\partial}{\partial x},\\
G_{15} &= u\left(v^2 + \frac{1}{v^2}\right)\frac{\partial}{\partial u} - v\left(u^2 - \frac{1}{v^2}\right)\frac{\partial}{\partial v}.
\end{aligned}$$

The system is obviously linearisable, as the fifteen-element algebra is $\mathfrak{sl}(4,\mathbb{R})$, the algebra of the two-dimensional free particle with equation of motion

$$\mathbf{r}'' = 0, \qquad \mathbf{r} \in \mathbb{R}^2.$$

Note that $\mathfrak{sl}(4,\mathbb{R})$ is not a prerequisite for linearisation. This is because the system

$$\mathbf{r}'' = A\mathbf{r}, \qquad \mathbf{r} \in \mathbb{R}^2,$$

need not possess $\mathfrak{sl}(4,\mathbb{R})$ symmetry.

From the symmetries, one looks to see if the algebra $\mathcal{N}$ is present. Recall that $\mathcal{N}$ has H, G_i, X_i such that

$$[H, G_i] = 0, \qquad [G_i, G_j] = 0,$$

and

$$[G_i, X_j] = 0, \qquad [H, X_i] = G_i.$$

In standard form,

$$H = \frac{\partial}{\partial x}, \qquad G_i = \frac{\partial}{\partial u_i}, \qquad X_i = x\frac{\partial}{\partial u_i}.$$

Noting that

$$[G_2, G_3] = 0, \qquad [G_2, G_6] = G_3, \qquad [G_3, G_6] = 0,$$

and

$$[G_2, G_{10}] = 0, \qquad [G_2, G_{12}] = G_{10}, \qquad [G_{10}, G_{12}] = 0,$$

we check to see if this is the algebra $\mathcal{N}$:

$$\begin{aligned}
[G_3, G_{10}] &= \left[\frac{1}{v}\frac{\partial}{\partial u} + \frac{1}{u}\frac{\partial}{\partial v}, v\frac{\partial}{\partial u} - \frac{v^2}{u}\frac{\partial}{\partial v}\right] \\
&= \left(\frac{1}{u} - \frac{1}{u}\right)\frac{\partial}{\partial u} + \left(\frac{v}{u^2} - \frac{2v}{u^2} + \frac{v}{u^2}\right)\frac{\partial}{\partial v} \\
&= 0.
\end{aligned}$$

Likewise,

$$[G_6, G_{10}] = 0, \qquad [G_6, G_{12}] = 0, \qquad [G_3, G_{12}] = 0.$$

Thus we do have $\mathcal{N}$. Identify

$$\begin{aligned}
H &= \frac{\partial}{\partial x}, \\
G_1 &= \frac{1}{v}\frac{\partial}{\partial u} + \frac{1}{u}\frac{\partial}{\partial v}, \\
G_2 &= v\frac{\partial}{\partial u} - \frac{v^2}{u}\frac{\partial}{\partial v}, \\
X_1 &= xG_1, \\
X_2 &= xG_2.
\end{aligned}$$

Seek a transformation from x, u, v to X, U, V such that

$$H = \frac{\partial}{\partial X}, \qquad G_1 = \frac{\partial}{\partial U}, \qquad G_2 = \frac{\partial}{\partial V},$$

$$X_1 = X\frac{\partial}{\partial U}, \qquad X_2 = X\frac{\partial}{\partial V}.$$

Let

$$X = x, \qquad U = F(u,v), \qquad V = G(u,v),$$

having already realised that x is an appropriate variable.
The derivatives transform according to

$$\frac{\partial}{\partial u} \longrightarrow \frac{\partial F}{\partial u}\frac{\partial}{\partial U} + \frac{\partial G}{\partial u}\frac{\partial}{\partial V},$$

and

$$\frac{\partial}{\partial v} \longrightarrow \frac{\partial F}{\partial v}\frac{\partial}{\partial U} + \frac{\partial G}{\partial v}\frac{\partial}{\partial V}.$$

Thus G_1 gives

$$\frac{1}{v}\frac{\partial F}{\partial u} + \frac{1}{u}\frac{\partial F}{\partial v} = 1, \tag{i}$$

and

$$\frac{1}{v}\frac{\partial G}{\partial u} + \frac{1}{u}\frac{\partial G}{\partial v} = 0. \tag{ii}$$

From G_2 we obtain

$$v\frac{\partial F}{\partial u} - \frac{v^2}{u}\frac{\partial F}{\partial v} = 0, \tag{iii}$$

and

$$v\frac{\partial G}{\partial u} - \frac{v^2}{u}\frac{\partial G}{\partial v} = 1. \tag{iv}$$

For (ii),

$$\frac{du}{1/v} = \frac{dv}{1/u}, \qquad w_1 = \frac{u}{v},$$

so

$$G = g\left(\frac{u}{v}\right).$$

In (iv),

$$v\frac{1}{v}g' - \frac{v^2}{u}\left(-\frac{u}{v^2}\right)g' = 1,$$

so

$$g' = \frac{1}{2},$$

and

$$G = \frac{1}{2}\frac{u}{v},$$

to within a forgettable constant.
For (i),

$$\frac{du}{1/v} = \frac{dv}{1/u} = \frac{dF}{1}.$$

The first two terms give

$$w_1 = \frac{u}{v}.$$

The first and third terms give

$$u\,du = dF.$$

Hence

$$w_2 = \frac{1}{2}uv - F,$$

and

$$F = \frac{1}{2}uv + f\left(\frac{u}{v}\right).$$

In (iii),

$$v\left(\frac{1}{2}v + \frac{1}{v}f'\right) - \frac{v^2}{u}\left(\frac{1}{2}u - \frac{u}{v^2}f'\right) = 0.$$

Therefore

$$2f' = 0,$$

so $f' = 0$, to within a forgettable constant, and

$$F = \frac{1}{2}uv.$$

Thus

$$V = \frac{1}{2}\frac{u}{v}, \qquad U = \frac{1}{2}uv.$$

$$\begin{aligned}
U' &= \frac{1}{2}u'v + \frac{1}{2}uv', \\
U'' &= \frac{1}{2}u''v + u'v' + \frac{1}{2}uv'' \\
&= \frac{1}{2}\left[-\frac{u(v')^2}{v^2}\right]v + u'v' + \frac{1}{2}u\left[\frac{(v')^2}{v} - \frac{2u'v'}{u}\right] \\
&= 0, \\
V'' &= 0.
\end{aligned}$$

Therefore

$$
\begin{aligned}
U &= A_0 + A_1 x, \\
V &= B_0 + B_1 x,
\end{aligned}
$$

giving

$$
\begin{aligned}
u^2 &= 4(A_0 + A_1 x)(B_0 + B_1 x), \\
v^2 &= \frac{A_0 + A_1 x}{B_0 + B_1 x},
\end{aligned}
$$

which is the general solution since there are four arbitrary constants.

1 LIE THEORY OF EXTENDED GROUP

In this chapter we outline the salient features of Lie theory which are of direct relevance to the following chapters. Other topics which enter naturally are discussed as the occasion arises.

1.1 Ordinary Differential Equations

In the study of the Lie theory of extended group, relevant to single second-order ordinary differential equations, a one-parameter point transformation acts on solution curves in (t, q) space and transforms solution curves into solution curves which possess the same value of the associated invariant (Lutzky 1979). This is equivalent to the requirement that the equation be form-invariant under the transformation. The set of all such one-parameter transformations for a particular differential equation forms a group. For such a monoparametric transformation group, the group operator is

$$G(t,q) = \xi(t,q)\frac{\partial}{\partial t} + \eta(t,q)\frac{\partial}{\partial q}. \tag{1.1}$$

For a function involving the nth ($n = 1, 2$ in the present study) derivative, the n-times extended group operator is given by

$$G^{(n)} = G^{(n-1)} + \eta^{(i)}\frac{\partial}{\partial q^{(i)}}, \tag{1.2}$$

where

$$\begin{aligned} \eta^{(k)} &= \frac{d\eta^{(k-1)}}{dt} - q^{(k)}\frac{d\xi}{dt}, \qquad k = 1, 2, \\ \frac{d}{dt} &= \frac{\partial}{\partial t} + \dot{q}\frac{\partial}{\partial q} + \ddot{q}\frac{\partial}{\partial \dot{q}}. \end{aligned} \tag{1.3}$$

The finite transformations of the group can be obtained by exponentiation of the group operator

$$\bar{t} = (\exp \alpha G)t, \qquad \bar{q} = (\exp \alpha G)q, \tag{1.4}$$

where α is the group parameter, or by integration of the system of differential equations

$$\frac{d\bar{t}}{\xi(\bar{t},\bar{q})} = \frac{d\bar{q}}{\eta(\bar{t},\bar{q})} = d\alpha, \tag{1.5}$$

subject to the initial conditions

$$\bar{t} = t, \qquad \bar{q} = q \qquad \text{when} \qquad \alpha = 0.$$

We obtain the infinitesimal transformation in (t, q) space by using (1.4). Employing the infinitesimal notation $\delta\alpha$ for the value of α in the immediate neighbourhood of $\alpha = 0$ and neglecting $O(\delta\alpha^2)$ terms, we can write (1.4) as

$$\bar{t} = t + \xi\,\delta\alpha, \qquad \bar{q} = q + \eta\,\delta\alpha. \tag{1.6}$$

The induced variations in the higher derivatives are likewise given by

$$\bar{q}' = \dot{q} + \eta^{(1)}\delta\alpha, \qquad \bar{q}'' = \ddot{q} + \eta^{(2)}\delta\alpha, \tag{1.7}$$

where $'$ denotes d/dt. This is easily verified by utilising the finite transformations of the first and second extended group

$$\bar{q}' = (\exp\alpha G^{(1)})\dot{q}, \qquad \bar{q}'' = (\exp\alpha G^{(2)})\ddot{q}. \tag{1.8}$$

For a general second-order differential equation

$$N(\ddot{q}, \dot{q}, q, t) = 0, \tag{1.9}$$

the second extended operator $G^{(2)}$ is used. An operator G is said to be the generator of a one-parameter symmetry group for (1.9) if, whenever (1.9) is satisfied,

$$G^{(2)}N(\ddot{q}, \dot{q}, q, t) = 0,$$

or equivalently

$$\xi\frac{\partial N}{\partial t} + \eta\frac{\partial N}{\partial q} + \eta^{(1)}\frac{\partial N}{\partial \dot{q}} + \eta^{(2)}\frac{\partial N}{\partial \ddot{q}} = 0. \tag{1.10}$$

Writing

$$N = \ddot{q} - M(\dot{q}, q, t), \tag{1.11}$$

(1.10) becomes

$$\eta^{(2)} - G^{(1)}M = 0. \tag{1.12}$$

Generally, a vector field

$$Y = \xi(t, q, \dot{q})\frac{\partial}{\partial t} + \eta(t, q, \dot{q})\frac{\partial}{\partial q} + \zeta(t, q, \dot{q})\frac{\partial}{\partial \dot{q}}$$

is said to be a dynamical symmetry of

$$\Gamma = \frac{\partial}{\partial t} + \dot{q}\frac{\partial}{\partial q} + M\frac{\partial}{\partial \dot{q}}$$

if

$$\mathcal{L}_Y\Gamma = [Y,\Gamma] = g\Gamma, \tag{1.13}$$

for a suitable function g. In this interpretation, the flow of Y maps integral curves of Γ into integral curves, subject to a change in the parametrization along the integral curves. Equation (1.13) gives rise to the following conditions (see e.g. Sarlet and Cantrijn 1981)

$$\begin{aligned} \zeta &= \Gamma(\eta) - \dot{q}\,\Gamma(\xi), \\ \Gamma(\zeta) - M\Gamma(\xi) - Y(M) &= 0, \\ g &= -\Gamma(\xi), \end{aligned} \tag{1.14}$$

which in turn implies

$$\Gamma^2(\eta) - \dot{q}\,\Gamma^2(\xi) - 2M\Gamma(\xi) - Y(M) = 0. \tag{1.15}$$

This is precisely the criterion derived by Anderson and Davison (1974) in the context of generalization of the Lie theory. It is also of interest to note that (1.15) is identical to (1.12) when ξ and η, satisfying (1.15), are independent of the velocity. In view of this, we say that Y (actually $Y^{(0)} = \xi\,\partial/\partial t + \eta\,\partial/\partial q$) is a Lie point symmetry of Γ.

For given ξ and η, a first integral may be obtained by imposing the double requirement

$$G^{(1)}I(t,q,\dot{q}) = 0, \qquad \frac{d}{dt}I(t,q,\dot{q}) = 0. \tag{1.16}$$

The function $I(t,q,\dot{q})$ satisfying (1.16a) is said to be an invariant of the first extended group. Equation (1.16a) gives rise to the following system of ordinary differential equations

$$\frac{dt}{\xi} = \frac{dq}{\eta} = \frac{d\dot{q}}{\eta^{(1)}}, \tag{1.17}$$

the solution of which produces two characteristics $u(t,q)$ and $v(t,q,\dot{q})$, referred to as the group invariant and the first order differential invariant respectively. Thus an arbitrary function $F(u,v)$ of u and v is the solution of (1.16a). Invoking (1.16b) leads to

$$\frac{\partial F}{\partial u}\dot{u} + \frac{\partial F}{\partial v}\dot{v} = 0,$$

where the dot denotes d/dt. Clearly we have

$$\frac{dv}{du} = \frac{\dot{v}}{\dot{u}}. \tag{1.18}$$

If M (as in (1.11)) is known, the right hand side of (1.18) can be written as a function of u and v, say $G(u,v)$, and we have

$$\frac{dv}{du} = G(u,v). \tag{1.19}$$

A first integral of the differential equation (1.11) is then obtained by solving (1.19).

The preceding discussion also shows how the knowledge of a point symmetry enables the reduction of a given second-order differential equation to a first-order equation, namely (1.19). Furthermore, the general second-order equation invariant under a given group can be expressed as

$$\frac{\dfrac{\partial v}{\partial t} + \dot{q}\dfrac{\partial v}{\partial q} + \ddot{q}\dfrac{\partial v}{\partial \dot{q}}}{\dfrac{\partial u}{\partial t} + \dot{q}\dfrac{\partial u}{\partial q}} = H\Big(u(t,q), v(t,q,\dot{q})\Big), \tag{1.20}$$

where H is an arbitrary function and the left hand side is the quotient $\dot{v}/\dot{u}$ of (1.19).

The following remark is important: If a first integral I (corresponding to G) is obtained by integration (in the above sense (1.16)), then it is possible to generate other first integrals and from these further first integrals until all operations $X^{(1)}I$ become meaningless, where $X^{(1)}$ is the first extension of $X \neq G$ belonging to the set of given operators. It is a simple matter to verify this. Indeed

$$\Gamma(I) = 0 \qquad \text{and} \qquad [X^{(1)}, \Gamma] = g\Gamma$$

implies

$$\Gamma(X^{(1)}I) = [\Gamma, X^{(1)}]I + X^{(1)}(\Gamma(I)) = 0,$$

so that $X^{(1)}I$ is potentially another first integral.

It is often desirable to introduce new coordinates $Q = F(t,q)$, $T = G(t,q)$ (frequently written as $Q = Q(t,q)$, $T = T(t,q)$ when no confusion arises) in which the generator (1.1) of the group (1.4) appears as a generator of time or space translation (or the generator of any other suitable group). In this respect we have

$$\begin{aligned} \bar{G}(t,q) &= (GT)\frac{\partial}{\partial T} + (GQ)\frac{\partial}{\partial Q} \\ &= \bar{\xi}(T,Q)\frac{\partial}{\partial T} + \bar{\eta}(T,Q)\frac{\partial}{\partial Q}. \end{aligned} \tag{1.21}$$

Thus

$$\begin{aligned} \xi(t,q)\frac{\partial T}{\partial t} + \eta(t,q)\frac{\partial T}{\partial q} &= \bar{\xi}(T,Q), \\ \xi(t,q)\frac{\partial Q}{\partial t} + \eta(t,q)\frac{\partial Q}{\partial q} &= \bar{\eta}(T,Q), \end{aligned} \tag{1.22}$$

the solution of which explicitly yields the new coordinates Q and T. If the group generated by (1.21) is one of translation, then Q and T are called canonical coordinates. This will be the case when either $\bar{\xi} = 1$ and $\bar{\eta} = 0$ (time-translation) or $\bar{\xi} = 0$ and $\bar{\eta} = 1$ (space-translation).

More details on the Lie theory of extended groups may be found in Dickson (1924) and Bluman and Cole (1974).

1.2 Lie Algebra

A Lie algebra consists of a vector space L over a field F, together with a binary operation of commutation [,] defined on L such that the following axioms are satisfied:

(a) bilinearity: for any $u, v, w \in L$ and $a, b \in F$

$$[au + bv, w] = a[u, w] + b[v, w],$$
$$[u, av + bw] = a[u, v] + b[u, w];$$

(b) antisymmetry: for any $u, v \in L$

$$[u, v] = -[v, u];$$

(c) the Jacobi identity: for any $u, v, w \in L$

$$[[u, v], w] + [[v, w], u] + [[w, u], v] = 0.$$

We shall (somewhat incorrectly) speak of the Lie algebra L, where we take $F = \mathbb{R}$. For our purpose, we define the binary operation of commutation on a Lie algebra L of operators as the commutator

$$[X, Y] = XY - YX \qquad \text{for any } X, Y \in L. \tag{1.23}$$

If a differential equation admits the operators X and Y (in the sense mentioned previously), then it also admits their commutator $[X, Y]$ (Ovsiannikov 1978). Consequently the set of all operators admitted by a given differential equation generates a Lie algebra. The largest admitted Lie algebra is called the full Lie algebra of the equation. We encounter, in this work, only finite-dimensional Lie algebras of dimensionality r, where $r \leq 8$. Thus it is usual to represent a finite-dimensional Lie algebra L by a table of commutators, i.e., by an $r \times r$ matrix in which the commutator $[G_i, G_j]$ $(i, j = 1, r$ where $\{G_k\}$ is the basis) is placed at the intersection of the ith row and jth column. The resulting matrix is antisymmetric and it is necessary to calculate only its upper half (see e.g. Mahomed and Leach 1985). Therefore we can write

$$[G_i, G_j] = C^k{}_{ij} G_k, \tag{1.24}$$

where the numbers $C^k{}_{ij}$ are called the structure constants of the Lie algebra L with respect to the basis $\{G_k\}$. These numbers are antisymmetric relative to the lower indices ($C^k{}_{ij} = -C^k{}_{ji}$) and satisfy the Jacobi identity

$$C^i{}_{jk} C^k{}_{lm} + C^i{}_{lk} C^k{}_{mj} + C^i{}_{mk} C^k{}_{jl} = 0.$$

The above properties are useful in constructing a Lie algebra.

It is clear from the relation

$$[G_i^{(n)}, G_j^{(n)}] = [G_i, G_j]^{(n)}, \qquad n \in \mathbb{N}, \tag{1.25}$$

that the n-times extended operators generate an r-dimensional Lie algebra, denoted $L^{(n)}$. Moreover $L^{(n)}$ has the same structure constants as L when referred to basis $\{G_k^{(n)}\}$,

$$[G_i^{(n)}, G_j^{(n)}] = C^k{}_{ij} G_k^{(n)}. \tag{1.26}$$

The classification of real low-dimensional Lie algebras was initiated by Lie (Lie 1891). It still engages scores of specialists (see e.g. Patera et al.). For many of them, the interest in studying the classification problem lies in its usefulness in many physical applications. Indeed, the operators admitted by a second-order differential equation generate a finite-dimensional Lie algebra of dimension at most eight. This is a direct consequence of Lie's counting theorem for second-order equations (Anderson and Davison 1974). Presently we concern ourselves with differential equations which admit two-dimensional Lie algebras of operators; deferring discussion on higher dimensional algebras to Chapter 5. There are two Lie algebras of dimension two, one abelian and one solvable (Barut and Raczka 1980):

$$[G_1, G_2] = 0, \qquad [G_1, G_2] = G_1. \tag{1.27}$$

Lie showed that second-order ordinary differential equations possessing two generators of symmetry have four canonical forms (Lie 1891). They are, with their associated generators in canonical form, given in the table below.

It is observed that there are two canonical forms for each of the Lie algebras (1.27a) and (1.27b).

Lie deduced that if an equation admits a two-dimensional Lie algebra with operators G_1, G_2 satisfying $G_2 = \rho(t,q) G_1$ (i.e. type 2 or 4) then it is linearizable. This is evident from the above table. Moreover a theorem of Lie states that every linear equation is reducible to the free particle equation. Hence an equation of type 2 or 4 can be transformed to the free particle equation and accordingly admits six additional operators.

REMARK: If a second-order equation admits the operators G_1 and G_2 then it follows from (1.24) that

$$[G_1, G_2] = aG_1 + bG_2, \qquad a, b \in \mathbb{R}. \tag{1.28}$$

For a and b both zero, G_1 and G_2 commute. However, when at least one of a or b is nonzero, we choose the basis $\{V_1, V_2\}$ so that $[V_1, V_2] = V_1$. This is easily done as follows: if $a \neq 0$ and $b \neq 0$ or $a \neq 0$ and $b = 0$, we introduce the basis $\{V_1 = G_1 + (b/a)G_2, V_2 = (1/a)G_2\}$. In the case $a = 0$ and $b \neq 0$, we choose $\{V_1 = G_2, V_2 = -(1/b)G_1\}$. The original equation admits the operators V_1 and V_2 since they are merely linear combinations of G_1 and G_2.

The contents of Table 1 are treated in greater detail in the following chapters. Various new features emerge.

For readable accounts on Lie algebras, the interested reader is referred to Wybourne (1974), Gilmore (1974) and Ovsiannikov (1978).

1.3 The Free Particle

The free particle, being the simplest system, is regarded as a paradigm of dynamical systems. It has been discussed by various authors (Lie 1891, Anderson and Davison 1974). Certain features of the problem, however, have received attention only recently (Mahomed and Leach 1985). Conventionally the symmetry generators of a one-dimensional system are used to determine the first integrals associated with the system. In the last cited reference, the reverse procedure was adopted for the free particle.

The one-dimensional free particle has equation

$$\ddot{q} = 0, \tag{1.29}$$

where the dot denotes d/dt, with Hamiltonian

$$H = \frac{1}{2}p^2, \qquad p = \dot{q}. \tag{1.30}$$

The two first integrals for (1.29) are easily seen to be

$$\begin{aligned} I_1 &= p, \\ I_2 &= q - tp. \end{aligned} \tag{1.31}$$

and we include their quotient

$$I_3 = \frac{q - tp}{p}, \tag{1.32}$$

since this has been shown by Leach (1980) to be relevant from the corresponding integral for the simple harmonic oscillator.

We seek the set of symmetry generators with which each of I_1, I_2, I_3 is associated.

ξ and η of the generator

$$G = \xi(t,q)\frac{\partial}{\partial t} + \eta(t,q)\frac{\partial}{\partial q} + \zeta(t,q,p)\frac{\partial}{\partial p} \tag{1.33}$$

are determined by the following equations

$$\zeta\frac{\partial I}{\partial p} + \eta\frac{\partial I}{\partial q} + \xi\frac{\partial I}{\partial t} = 0, \tag{1.34}$$

$$\eta^{(1)} - \zeta\frac{\partial^2 H}{\partial p^2} - \eta\frac{\partial^2 H}{\partial q\partial p} - \xi\frac{\partial^2 H}{\partial t\partial p} = 0, \tag{1.35}$$

where

$$\eta^{(1)} = \dot{\eta} - \dot{\xi}\frac{\partial H}{\partial p}, \tag{1.36}$$

by eliminating ζ between (1.34) and (1.35) and insisting that ξ and η be independent of p (Leach 1980).

Thus for the first integral $I_1 = p$ we obtain

$$\frac{d\eta}{dt} - p\frac{d\xi}{dt} = 0, \tag{1.37}$$

where

$$\frac{d}{dt} = \frac{\partial}{\partial t} + p\frac{\partial}{\partial q}.$$

(1.37) yields a partial differential equation in which the terms are grouped together in powers of p. By equating coefficients of separate powers of p to zero we obtain

$$\begin{aligned} p^2: &\qquad \xi = a(t), \\ p^1: &\qquad \eta = \dot{a}q + b(t), \end{aligned} \tag{1.38}$$

and

$$\begin{aligned} p^0: &\qquad a = At + B, \\ &\qquad b = C, \end{aligned} \tag{1.39}$$

where A, B and C are constants.

Therefore the triplet of generators with which I_1 is associated is

$$G_1 = t\frac{\partial}{\partial t} + q\frac{\partial}{\partial q}, \quad G_2 = \frac{\partial}{\partial t}, \qquad G_3 = \frac{\partial}{\partial q}. \tag{1.40}$$

In like manner, for I_2, we obtain

$$G_4 = t\frac{\partial}{\partial t}, \quad G_5 = t^2\frac{\partial}{\partial t} + tq\frac{\partial}{\partial q}, \qquad G_6 = t\frac{\partial}{\partial q}. \tag{1.41}$$

Similarly, for I_3,

$$G_7 = tq\frac{\partial}{\partial t} + q^2\frac{\partial}{\partial q}, \quad G_8 = q\frac{\partial}{\partial t}, \qquad G_9 = q\frac{\partial}{\partial q}. \tag{1.42}$$

It is observed that G_1, G_4 and G_9 are linearly dependent. Indeed we have

$$G_1 = G_4 + G_9. \tag{1.43}$$

We now compare the standard generators (Lie 1891) of the free particle with those obtained here using the reverse procedure. In order to do this, we need to determine the usual free particle generators using the (direct) Lie method of extended group (see Section 1.1).

An operator of the form (1.1) will be a symmetry generator for (1.29) if and only if ξ and η satisfy (invariance of (1.29) under $G^{(2)}$)

$$\eta^{(2)} = 0, \tag{1.44}$$

which, when different powers of $\dot{q}$ are treated as linearly independent (as they are, since we have a second-order equation) yields, on equating the coefficients of separate powers of $\dot{q}$ to zero (bearing in mind that ξ and η are functions of t and q only), for the third and second powers

$$\dot{q}^3: \qquad \xi = a(t)q + b(t), \tag{1.45}$$

and

$$\dot{q}^2: \qquad \eta = \dot{a}q^2 + c(t)q + d(t), \tag{1.46}$$

where the time-dependent functions a, b, c, and d are solutions of

$$\begin{aligned} \dot{q}^1: &\qquad \ddot{a} = 0, \quad 2\dot{c} - \ddot{b} = 0, \\ \dot{q}^0: &\qquad \ddot{c} = 0, \qquad\quad \ddot{d} = 0. \end{aligned} \tag{1.47}$$

We note that this system of equations has eight linearly independent solutions. Hence (1.29) has eight generators of symmetry which correspond identically to the generators $G_2, G_3, \ldots, G_9$ obtained via the reverse procedure.

To illustrate how one obtains a first integral for a given generator, we solve equations (1.16) for G_4. The first of (1.16) leads to (cf. (1.17))

$$\frac{dt}{t} = \frac{d\dot{q}}{-\dot{q}} = \frac{dq}{0}, \tag{1.48}$$

the solution of which produces the characteristics

$$u = q, \qquad v = t\dot{q}.$$

Applying equation (1.18) we then obtain

$$\frac{dv}{du} = 1. \tag{1.49}$$

Whence the first integral associated with G_4 is

$$I_2 = q - t\dot{q},$$

as expected. In the same way one can obtain the first integrals I_1 and I_2 corresponding to the generators given previously.

Of great interest to us are the Lie algebraic properties of the nine generators which in some sense are simpler than those for the usual eight as we shall see below. The commutation relations between the G_i's are given in the following table.

The above table cannot be regarded in the usual sense since there is a linear dependence relation between G_1, G_4 and G_9. However, there are certain nice features that can be noted as we shall soon discuss. Moreover, the conventional table is still available if we disregard the entries $[G_1, G_j]$ and $[G_i, G_1]$ $(i, j = 1, 9)$.

Each of the triplets of generators $\{G_1, G_2, G_3\}$, $\{G_4, G_5, G_6\}$ and $\{G_7, G_8, G_9\}$, associated with I_1, I_2 and I_3 respectively, forms a subalgebra. This is evident from the table (see diagonal blocks). Furthermore, each of these subalgebras can be written in the form

$$[X_1, X_2] = 0, \qquad [X_2, X_3] = X_2, \qquad [X_1, X_3] = X_1. \tag{1.50}$$

This can be accomplished by changes in sign of G_4 and G_9. With these changes in sign, the three subalgebras become isomorphic to each other.

Let us recall the three triplets of generators (1.40), (1.41) and (1.42). We note that none of them contain operators that are connected to each other i.e. we do not have operators X_1, X_2, X_3 such that

$$X_1 = \rho(t,q)X_2 \qquad \text{and} \qquad X_3 = \psi(t,q)X_2$$

for suitable functions ρ and ψ. It follows therefore, from (1.50) that the triplets of generators are equivalent to each other under point transformation. Indeed, by inspection we observe that the I_2-generators are equivalent to the I_3-generators under the interchange transformation

$$T = q, \qquad Q = t. \tag{1.51}$$

The I_1-generators are form-invariant under this transformation. By a straightforward calculation we can show that the I_2-generators transform into the I_1-generators via the point transformation

$$T = -\frac{1}{t}, \qquad Q = \frac{q}{t}. \tag{1.52}$$

Hence using (1.51) together with (1.52) we can deduce that the transformation which reduces the I_3-generators to the I_1-generators is

$$T = -\frac{1}{q}, \qquad Q = \frac{t}{q}. \tag{1.53}$$

Clearly the above transformations leave the free particle equation invariant.

We do not wish to labour the point in discussing the three-dimensional algebra (1.50) any further since a more general treatment pertaining to it is given in Chapter 5. However, we do remark that the two functionally independent first integrals (any two of I_1, I_2, I_3), sufficient for obtaining the complete solution of the free particle equation, each has associated a triplet of operators with isomorphic algebras. The complete solution is essentially provided by any two generators, each belonging to a different class of generators (any two of I_1-generators, I_2-generators, I_3-generators).

We wish to classify the Lie algebra and identify the full symmetry group for the free particle equation (1.29). The metric tensor of the Lie algebra is given by

$$g_{ij} = C^m{}_{ik}C^k{}_{jm}, \tag{1.54}$$

where $C^m{}_{ik}$ are the structure constants. Cartan's criterion for semi-simplicity requires that the determinant of g_{ij} be non-vanishing (see e.g. Gilmore 1974). The metric tensor can be shown to satisfy this requirement. Indeed, the following eight linearly independent linear combinations of the nine operators (1.40), (1.41) and (1.42) are a basis for a Lie algebra having a diagonal metric tensor:

$$
\begin{aligned}
Y_1 &= -(1+t^2)\frac{\partial}{\partial t} - tq\frac{\partial}{\partial q},\\
Y_2 &= (1-t^2)\frac{\partial}{\partial t} - tq\frac{\partial}{\partial q},\\
Y_3 &= tq\frac{\partial}{\partial t} + (1+q^2)\frac{\partial}{\partial q},\\
Y_4 &= tq\frac{\partial}{\partial t} + (q^2-1)\frac{\partial}{\partial q},\\
Y_5 &= -q\frac{\partial}{\partial t} + t\frac{\partial}{\partial q},\\
Y_6 &= -t\frac{\partial}{\partial t} - q\frac{\partial}{\partial q},\\
Y_7 &= t\frac{\partial}{\partial t} - q\frac{\partial}{\partial q},\\
Y_8 &= -q\frac{\partial}{\partial t} - t\frac{\partial}{\partial q}.
\end{aligned}
\tag{1.55}
$$

This follows from the work of Wulfman and Wybourne (1974) on the harmonic oscillator. The Lie algebra of the operators Y_i is isomorphic to the Lie algebra of the harmonic oscillator operators which has diagonal metric. Moreover, the metric tensor is indefinite and so the Lie group generated by the Lie algebra is non-compact. It should also be noted that $\{Y_1, Y_3, Y_5\}$ constitutes a compact subalgebra, associated with a negative definite metric $g_{ij} = -2\delta_{ij}$, which generates a compact Lie group $SO(3)$.

We can select linear combinations of the Y_i so that the Lie algebra is cast into the Cartan-Weyl standard form (Wybourne 1974), thus leading to its identification as a non-compact form of Cartan's A_2 algebra. The A_2 algebra can only generate the three Lie groups $SU(3)$, $SU(2,1)$ or $SL(3,\mathbb{R})$. Since only the last is both non-compact and in possession of an $SO(3)$ subgroup, we identify the full symmetry group of the free particle equation (1.29) as the Lie group $SL(3,\mathbb{R})$.

We may regard the free particle equation (1.29) as a canonical form for dynamical systems, linear as well as nonlinear, possessing the full symmetry group $SL(3,\mathbb{R})$. Thus we can map any nonlinear differential equation having $SL(3,\mathbb{R})$ symmetry into the free particle equation in a one-one manner. In view of this, the invariance properties of the free particle equation are injected into the nonlinear equation.

Linear second-order ODEs

Factorisation technique

For example,

$$y'' + y = 0.$$

$$(D^2 + 1)y = 0,$$
$$(D + i)(D - i)y = 0.$$

Let

$$(D - i)y = z.$$

$$(D + i)z = 0,$$

$$z' + iz = 0.$$

Integrating factor: e^{ix}.

$$(ze^{ix})' = 0,$$

$$ze^{ix} = -2iB.$$

$$z = -2iBe^{-ix}.$$

$$y' - iy = -2iBe^{-ix}.$$

Integrating factor: e^{-ix}.

$$(ye^{-ix})' = -2iBe^{-2ix}.$$

$$ye^{-ix} = A + Be^{-2ix},$$

$$y = Ae^{ix} + Be^{-ix},$$

or

$$y = C \sin x + D \cos x.$$

Equally could use

$$(D - i)(D + i)y = 0.$$

For a less trivial example consider

$$y'' - (x^2+1)y = 0.$$

$$(D+x)(D-x)y = 0.$$

Note this is not the same as

$$(D-x)(D+x) = D^2 - (x^2-1).$$

Let

$$z = (D-x)y.$$

$$(D+x)z = 0.$$

Integrating factor: $e^{x^2/2}$.

$$(ze^{x^2/2})' = 0,$$

$$z = Be^{-x^2/2}.$$

$$(D-x)y = Be^{-x^2/2}.$$

Integrating factor: $e^{-x^2/2}$.

$$y = Ae^{x^2/2} + Be^{x^2/2}\int e^{-x^2}\,dx.$$

Consider

$$y'' - f(x)y = 0$$

and the factorisation

$$(D+\alpha(x))(D+\beta(x)).$$

$$(D+\alpha)(D+\beta) = D^2 + (\alpha+\beta)D + \alpha\beta + \beta',$$

$$\beta = -\alpha, \qquad \alpha' + \alpha^2 = f.$$

where $\alpha' + \alpha^2 = f$ is Riccati's equation.

$$(D+\alpha)(D-\alpha)y = 0,$$

$$z = (D-\alpha)y.$$

$$z = Be^{\int \alpha\,dx},$$

leads to

$$y = Ae^{-\int \alpha\,dx} + Be^{-\int \alpha\,dx}\int e^{2\int \alpha\,dx}\,dx.$$

Existence of solutions

Initial value problem:

$$y'' - f(x)y = 0, \qquad y(0) = \alpha, \qquad y'(0) = \beta,$$

has a continuous solution with continuous derivative if $f(x)$ is continuous.
Boundary value problem:

$$y'' - f(x)y = 0,$$

$$\begin{aligned} \alpha_{11}y(a) + \alpha_{12}y'(a) + \beta_{11}y(b) + \beta_{12}y'(b) = 0, \\ \alpha_{21}y(a) + \alpha_{22}y'(a) + \beta_{21}y(b) + \beta_{22}y'(b) = 0. \end{aligned}$$

where the two conditions are independent. This may or may not have a non-trivial solution. For example,

$$y'' + y = 0, \qquad y(0) = 0, \qquad y'(\pi) = 0.$$

$$y = A\sin x + B\cos x,$$

$$y(0) = 0 \quad \Longrightarrow \quad B = 0,$$

and

$$y'(\pi) = 0 \quad \Longrightarrow \quad -A = 0.$$

There is no non-trivial solution.
However, consider

$$y'' + \lambda y = 0, \qquad y(0) = 0, \qquad y'(\pi) = 0.$$

For $\lambda = -\omega^2$,

$$y = A\sinh\omega x + B\cosh\omega x,$$

$$y(0) = 0 \quad \Longrightarrow \quad B = 0,$$

$$y'(\pi) = 0 \quad \Longrightarrow \quad A = 0.$$

For $\lambda = 0$,

$$y = Ax + B,$$

$$y(0) = 0 \quad \Longrightarrow \quad B = 0,$$

$$y'(\pi) = 0 \quad \Longrightarrow \quad A = 0.$$

For $\lambda = \omega^2$,

$$y = A \sin \omega x + B \cos \omega x.$$

$$y(0) = 0 \quad \Longrightarrow \quad B = 0,$$

$$y'(\pi) = 0 \quad \Longrightarrow \quad \omega A \cos(\omega\pi) = 0.$$

that is,

$$\cos \omega\pi = 0,$$

$$\omega = n + \frac{1}{2}.$$

Solution is

$$y = A \sin\left(n + \frac{1}{2}\right) x.$$

$n + \frac{1}{2}$ is called the eigenvalue and $\sin\left(n + \frac{1}{2}\right) x$ is called the eigenfunction, sometimes the eigenstate.

Standard form of the eigenvalue differential equation

Given

$$\frac{d}{dx}\left(f \frac{dy}{dx}\right) + gy + \lambda hy = 0,$$

where f, g, and h are functions of x, this can be transformed to the standard form

$$\frac{d^2Y}{dX^2} + F(X)Y + \lambda Y = 0$$

by the transformation

$$Y = (fh)^{1/4} y, \qquad dX = \left(\frac{h}{f}\right)^{1/2} dx.$$

f and h must be of the same sign and h/f must exist everywhere in the domain of applicability of the equation. The standard form is called the normal form.

Example

$$y'' - x^2 y + \lambda y = 0,$$

Write as

$$Hy = \lambda y, \qquad H = x^2 - D^2.$$

Let

$$a = x + D, \qquad a^\dagger = x - D.$$

Using $\alpha = x$ and $\alpha = -x$ gives

$$\begin{aligned}(x-D)(x+D)y &= (\lambda+1)y,\\ (x+D)(x-D)y &= (\lambda-1)y.\end{aligned}$$

$$\begin{aligned}[a, a^\dagger] &= (x+D)(x-D) - (x-D)(x+D)\\ &= x^2 + 1 - D^2 - (x^2 - 1 - D^2)\\ &= 2,\end{aligned}$$

$$\begin{aligned}[H, a] &= (x^2 - D^2)(x+D) - (x+D)(x^2 - D^2)\\ &= x^3 + x^2D - D^2x - D^3 - x^3 + xD^2 - Dx^2 + D^3\\ &= x^2D - D^2x + xD^2 - Dx^2\\ &= -2x - 2D\\ &= -2a,\end{aligned}$$

and

$$\begin{aligned}[H, a^\dagger] &= (x^2 - D^2)(x-D) - (x-D)(x^2 - D^2)\\ &= x^3 - x^2D - D^2x + D^3 - x^3 + xD^2 + Dx^2 - D^3\\ &= -x^2D - D^2x + xD^2 + Dx^2\\ &= 2x - 2D\\ &= 2a^\dagger.\end{aligned}$$

If we include the identity, we have the algebra with elements $H, a, a^\dagger, 1$ such that

$$[H, a] = -2a, \qquad [H, a^\dagger] = 2a^\dagger, \qquad [a, a^\dagger] = 2\,1, \qquad [H, 1] = 0,$$

etc.

Suppose ψ is a solution of $\psi'' + (\lambda - x^2)\psi = 0$ corresponding to the eigenvalue λ. Then, denoting it by ψ_λ, we have

$$H\psi_\lambda = \lambda\psi_\lambda.$$

Then

$$\begin{aligned}H(a\psi_\lambda) &= (Ha)\psi_\lambda\\ &= ([H, a] + aH)\psi_\lambda\\ &= -2a\psi_\lambda + a\lambda\psi_\lambda\\ &= (\lambda - 2)a\psi_\lambda,\end{aligned}$$

that is, $a\psi_\lambda$ is the eigenfunction corresponding to the eigenvalue $\lambda - 2$. Likewise with $a^\dagger$,

$$\begin{aligned}H(a^\dagger\psi_\lambda) &= (Ha^\dagger)\psi_\lambda\\ &= ([H, a^\dagger] + a^\dagger H)\psi_\lambda\\ &= 2a^\dagger\psi_\lambda + a^\dagger\lambda\psi_\lambda\\ &= (\lambda + 2)a^\dagger\psi_\lambda,\end{aligned}$$

that is, $a^\dagger\psi_\lambda$ is the eigenfunction corresponding to eigenvalue $\lambda+2$. a and $a^\dagger$ are called ladder operators. a is also called a lowering operator and $a^\dagger$ a raising operator.

Assume that ψ_λ is L^2 over $(-\infty,\infty)$, that is,

$$\|\psi_\lambda\|^2 = \int_{-\infty}^{\infty} |\psi_\lambda|^2\,dx < \infty.$$

$$\begin{aligned}(\psi_\lambda, x\psi_\lambda) &= \int_{-\infty}^{\infty} \psi_\lambda^* x\psi_\lambda\,dx \\ &= \int_{-\infty}^{\infty} (x\psi_\lambda)^*\psi_\lambda\,dx \\ &= (x\psi_\lambda, \psi_\lambda),\end{aligned}$$

and

$$\begin{aligned}(\psi_\lambda, D\psi_\lambda) &= \int_{-\infty}^{\infty} \psi_\lambda^*\psi_\lambda'\,dx \\ &= [\psi_\lambda^*\psi_\lambda]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} (\psi_\lambda^*)'\psi_\lambda\,dx \\ &= -(D\psi_\lambda, \psi_\lambda),\end{aligned}$$

since D is a real operator.

Hence

$$\begin{aligned}(\psi, a\psi) &= (a^\dagger\psi, \psi), \\ (\psi, a^\dagger\psi) &= (a\psi, \psi).\end{aligned}$$

We can write

$$H = a^\dagger a + 1,$$

so

$$a^\dagger a = H - 1,$$

$$(\psi_\lambda, a^\dagger a\psi_\lambda) = (\psi_\lambda, (\lambda-1)\psi_\lambda) = (\lambda-1)\|\psi_\lambda\|^2 \geq 0.$$

where equality is attained only if $\lambda = 1$, since ψ_λ is assumed to be non-trivial (that is, not identically zero). However,

$$(\psi_\lambda, a^\dagger a\psi_\lambda) = (a\psi_\lambda, a\psi_\lambda) = \|a\psi_\lambda\|^2 = (\lambda-1)\|\psi_\lambda\|^2.$$

So if $\lambda = 1$, then $\|a\psi_\lambda\|^2 = 0$, that is,

$$a\psi_\lambda = 0.$$

that is,

$$(x+D)y = 0,$$

$$xy + y' = 0.$$

Integrating factor:

$$\left(ye^{x^2/2}\right)' = 0,$$

$$y = \text{constant}\, e^{-x^2/2}.$$

To obtain the constant K, impose a value on the norm, usually one:

$$1 = |K|^2 \int_{-\infty}^{\infty} e^{-x^2}\, dx = |K|^2 \sqrt{\pi},$$

so $K = \pi^{-1/4}$.
Call this solution

$$\psi_0 = e^{-x^2/2}$$

(forget about the constant).

$$a^\dagger \psi_0 = \left(x - \frac{d}{dx}\right) e^{-x^2/2} = 2xe^{-x^2/2},$$

$$\lambda = 1 + 2 = 3.$$

Call this ψ_1. Likewise,

$$\psi_2 = a^\dagger \psi_1 = (4x^2 - 2)e^{-x^2/2},$$

$$\lambda = 3 + 2 = 5.$$

In general,

$$\psi_n = (a^\dagger)^n \psi_0 = \left(x - \frac{d}{dx}\right)^n e^{-x^2/2}. \qquad (*)$$

and

$$\lambda_n = 2n + 1.$$

ψ_n has the form

$$\psi_n = H_n(x) e^{-x^2/2}. \qquad (**)$$

(Note: normalisation is neglected.)
where $H_n(x)$ is the nth Hermite polynomial. From $(*)$ and $(**)$ we obtain the Rodrigues formula for the Hermite polynomials:

$$H_n(x) = e^{x^2/2} \left(x - \frac{d}{dx}\right)^n e^{-x^2/2}.$$

Note: $a^\dagger a$ is an example of an Hermitian operator, for

$$\begin{aligned}(\phi, a^\dagger a\psi) &= \int_{-\infty}^{\infty} \phi^*(a^\dagger a\psi)\,dx \\ &= \int_{-\infty}^{\infty} (a\phi)^*(a\psi)\,dx \\ &= (a\phi, a\psi) \\ &= (a^\dagger a\phi, \psi).\end{aligned}$$

The eigenvalues of an Hermitian operator are real. Let O be an Hermitian operator:

$$O\psi = \lambda\psi, \qquad O\phi = \mu\phi,$$

then

$$\begin{aligned}(O\phi, \psi) &= \mu^*(\phi, \psi), \\ (\phi, O\psi) &= \lambda(\phi, \psi).\end{aligned}$$

Now $(\phi, O\psi) = (O\phi, \psi)$, hence

$$(\lambda - \mu^*)(\phi, \psi) = 0.$$

If $\phi = \psi$, then $(\lambda - \bar{\lambda})\|\psi\|^2 = 0$, that is, $\lambda = \bar{\lambda}$, so it is real. If $\phi \neq \psi$ and $\lambda \neq \mu$, then $(\lambda - \mu)(\phi, \psi) = 0$, that is, $(\phi, \psi) = 0$, that is, states are orthogonal.

Hence we have an infinite-dimensional function space spanned by the set of functions $\psi_n(x) = H_n(x)e^{-x^2/2}$, and these are orthogonal. This makes the Hermite polynomials a suitable basis for functions over $(-\infty, \infty)$. That is, one writes a function $f(x)$ as a Fourier–Hermite series

$$f(x) = \sum_{n=0}^{\infty} a_n H_n(x) e^{-x^2/2}.$$

Orthogonal polynomials

Orthogonality:

$$\int_a^b w(x) f_m(x) f_n(x)\,dx = h_m \delta_{mn},$$

where $w(x)$ is a weight function, h_m is some constant and δ_{mn} is the Kronecker delta. Often take $h_m = 1$, but sometimes other values are more convenient.

Differential equation

$$\left[g_2(x)D^2 + g_1(x)D + \lambda_n\right] f_n = 0, \qquad \lambda_n = \lambda(n).$$

Recurrence

$$f_{n+1} = (a_n + x b_n) f_n - c_n f_{n-1}.$$

Rodrigues' formula

$$f_n(x) = \frac{1}{e_n w(x)} \frac{d^n}{dx^n} \left\{ w(x) [g(x)]^n \right\}.$$

where e_n is a constant and $w(x)$ is the weight function (not always found).

Generating function

$$g(x,z) = \sum_{n=0}^{\infty} a_n f_n(x) z^n.$$

Differential relations

$$m_2(x)\frac{df_n}{dx} = m_1(x) f_n + m_0(x) f_{n-1}.$$

Integral representation

$$f_n(x) = \frac{p_0(x)}{2\pi i}\oint_\Gamma [p_1(z,x)]^n p_2(z,x)\,dz.$$

where Γ is a closed contour taken around some specified point in the positive sense.

Consider Hermite polynomials.

$$H_n'' - 2xH_n' + 2nH_n = 0,$$

$H_n(x)$ satisfies the above equation, and $K_n = H_n e^{-x^2/2}$ satisfies

$$K_n'' + (2n+1-x^2)K_n = 0.$$

Orthogonality already established.

Rodrigues' formula:

Had

$$H_n(x) = e^{x^2/2}\left(x - \frac{d}{dx}\right)^n e^{-x^2/2},$$

This is not in standard form. The standard form is

$$H_n(x) = e^{x^2}\left(-\frac{d}{dx}\right)^n e^{-x^2}.$$

Obviously correct for $n = 0$. Let $n = 1$:

$$\begin{aligned}
e^{x^2/2}\left(x - \frac{d}{dx}\right) e^{-x^2/2} &= e^{x^2/2}\left(x - \frac{d}{dx}\right)\left(e^{x^2/2}e^{-x^2}\right) \\
&= e^{x^2/2}\left[\left(x - \frac{d}{dx}\right)e^{x^2/2}\right]e^{-x^2} + e^{x^2}\left(-\frac{d}{dx}\right)e^{-x^2} \\
&= e^{x^2/2}\left[(x-x)e^{x^2/2}\right]e^{-x^2} + e^{x^2}\left(-\frac{d}{dx}\right)e^{-x^2} \\
&= e^{x^2}\left(-\frac{d}{dx}\right)e^{-x^2}.
\end{aligned}$$

That is, it is true for $n = 1$. Assume it is true for $n = k$. Then for $n = k+1$,

$$
\begin{aligned}
e^{x^2/2}\left(x-\frac{d}{dx}\right)^{k+1} e^{-x^2/2} &= e^{x^2/2}\left(x-\frac{d}{dx}\right) e^{-x^2/2}\left(-\frac{d}{dx}\right)^{k} e^{-x^2} \\
&= \left[e^{x^2/2}\left(x-\frac{d}{dx}\right) e^{x^2/2}\right]\left(-\frac{d}{dx}\right)^{k} e^{-x^2} \\
&= e^{x^2}\left(-\frac{d}{dx}\right)^{k+1} e^{-x^2}.
\end{aligned}
$$

Q.E.D.

Recurrence formula:

$$
\begin{aligned}
H_{n+1}(x) &= e^{x^2}\left(-\frac{d}{dx}\right)^{n+1} e^{-x^2} \\
&= e^{x^2}\left(-\frac{d}{dx}\right)^{n}\left(2xe^{-x^2}\right).
\end{aligned}
$$

Using Leibniz's rule,

$$
\begin{aligned}
H_{n+1}(x) = e^{x^2}\sum_{j=0}^{n}\binom{n}{j}&\left[\left(-\frac{d}{dx}\right)^{j}(2x)\right]\left[\left(-\frac{d}{dx}\right)^{n-j} e^{-x^2}\right] \\
&= e^{x^2}\binom{n}{0}(2x)\left(-\frac{d}{dx}\right)^{n} e^{-x^2} + e^{x^2}\binom{n}{1}(-2)\left(-\frac{d}{dx}\right)^{n-1} e^{-x^2} \\
&= 2xH_n(x) - 2nH_{n-1}(x).
\end{aligned}
$$

$$
H_{n+1} = 2xH_n - 2nH_{n-1},
$$

Differential recurrence relations:

$$
\begin{aligned}
\frac{dH_n}{dx} &= \frac{d}{dx}\left[e^{x^2}\left(-\frac{d}{dx}\right)^{n} e^{-x^2}\right] \\
&= 2xe^{x^2}\left(-\frac{d}{dx}\right)^{n} e^{-x^2} - e^{x^2}\left(-\frac{d}{dx}\right)^{n+1} e^{-x^2} \\
&= 2xH_n - H_{n+1} \\
&= 2xH_n - (2xH_n - 2nH_{n-1}) \\
&= 2nH_{n-1}.
\end{aligned}
$$

$$
H_n' = 2nH_{n-1}.
$$

Generating function:

$$
\begin{aligned}
f(x,z) &= \sum_{n=0}^{\infty} a_n H_n(x) z^n \\
&= e^{x^2}\left\{\sum_{n=0}^{\infty} a_n\left(-z\frac{\partial}{\partial x}\right)^{n}\right\} e^{-x^2}.
\end{aligned}
$$

Let $a_n = 1/n!$:

$$
\begin{aligned}
f(x,z) &= e^{x^2}\exp\left(-z\frac{\partial}{\partial x}\right)e^{-x^2} \\
&= e^{x^2}\exp\left\{-x^2+\left[-z\frac{\partial}{\partial x},-x^2\right]+\frac{1}{2!}\left[-z\frac{\partial}{\partial x},\left[-z\frac{\partial}{\partial x},-x^2\right]\right]+\cdots\right\} \\
&= e^{2xz-z^2},
\end{aligned}
$$

[This uses a Baker–Campbell–Hausdorff formula.]

$$\sum_{n=0}^{\infty} H_n(x)\frac{z^n}{n!} = e^{2xz-z^2}.$$

Integral representation:
we have

$$\frac{e^{2xz-z^2}}{z^{m+1}} = \sum_{n=0}^{\infty}\frac{1}{n!}H_n(x)z^{n-m-1}.$$

The right-hand side is now a Laurent series about $z=0$.
(A Laurent series $\sum_{n=-k}^{\infty} a_n z^n$, that is, negative powers.)
The coefficient of z^{-1} is the residue at $z=0$, and

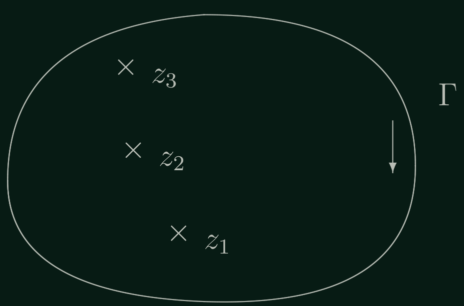
$$\oint_\Gamma \frac{e^{2xz-z^2}}{z^{m+1}}\,dz = 2\pi i\frac{H_m(x)}{m!},$$

if Γ encloses zero. The coefficient of z^{-1} is

$$\frac{1}{m!}H_m(x),$$

so

$$H_n(x) = \frac{n!}{2\pi i}\oint_\Gamma \frac{e^{2xz-z^2}}{z^{n+1}}\,dz.$$



$f(z)$ has poles at z_1, z_2 and z_3.
Near z_1 (say),

$$f(z) \sim (z-z_1)^{-m}g(z),$$

where $g(z_1)$ is finite.

$$\oint_\Gamma f(z)\,dz = 2\pi i\left[\mathrm{Res}(z_1)+\mathrm{Res}(z_2)+\mathrm{Res}(z_3)\right].$$

For a pole of order m,

$$\mathrm{Res}(z_1) = \lim_{z\to z_1}\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-z_1)^m f(z)\right].$$

Example in spherical polars:

$$\left[-\nabla^2 + f(r)\right]y = \lambda y.$$

$$\left[-\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) - \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + f(r)\right]y = \lambda y.$$

Let

$$y = R(r)\Theta(\theta)\Phi(\phi),$$

$$\begin{aligned}-\frac{1}{r^2R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{1}{r^2\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)\\ -\frac{1}{r^2\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2} + f(r) = \lambda.\end{aligned}$$

Multiplying by $r^2\sin^2\theta$,

$$-\frac{\sin^2\theta}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{\sin\theta}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) - \frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} + r^2\sin^2\theta f(r) = \lambda r^2\sin^2\theta.$$

Evidently $\Phi''/\Phi = k$. For $k=\alpha^2$,

$$\Phi = A\sinh\alpha\phi + B\cosh\alpha\phi,$$

but we want Φ to have period 2π so that y is a single-valued function, hence $A=B=0$. For $k=0$, $\Phi = A\phi + B$ and $A=0$. For $k=-\alpha^2$,

$$\Phi = A\sin\alpha\phi + B\cos\alpha\phi,$$

which has period 2π if $\alpha = m$, an integer. Hence

$$\frac{\Phi''}{\Phi} = -m^2.$$

$$-\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{m^2}{\sin^2\theta} + r^2 f(r) = \lambda r^2.$$

that is,

$$-\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{m^2}{\sin^2\theta} = \mu$$

and

$$-\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+r^2f(r)=\mu+\lambda r^2.$$

Two eigenvalue problems. The second depends upon the function $f(r)$, whereas the first is purely geometric. Notice that the first also depends upon the parameter m.

Return to the example after a theory break.

Standard form:

$$y''+r(x,m)y+\lambda y=0. \tag{1}$$

Note: include m as motivated by example.

Definition of factorisation:

Equation (1) may be factorised if it can be replaced by each of the two equations

$$H_{m+1}H^{\dagger}_{m+1}y(\lambda,m)=[\lambda-L(m+1)]y(\lambda,m). \tag{2}$$

$$H^{\dagger}_{m}H_{m}y(\lambda,m)=[\lambda-L(m)]y(\lambda,m). \tag{3}$$

where

$$H_m=k(x,m)+D, \qquad H^{\dagger}_m=k(x,m)-D.$$

and L is some function of m.

Theorem. If $y(\lambda,m)$ is a solution of equation (1), then

$$y(\lambda,m+1)=H^{\dagger}_{m+1}y(\lambda,m),$$

and

$$y(\lambda,m-1)=H_my(\lambda,m)$$

are corresponding to the same λ, but different m.

Proof. Multiply equation (2) by $H^{\dagger}_{m+1}$:

$$\begin{aligned}H^{\dagger}_{m+1}(H_{m+1}H^{\dagger}_{m+1}y)&=[\lambda-L(m+1)]H^{\dagger}_{m+1}y,\\(H^{\dagger}_{m+1}H_{m+1})(H^{\dagger}_{m+1}y)&=[\lambda-L(m+1)](H^{\dagger}_{m+1}y).\end{aligned}$$

From equation (3),

$$H^{\dagger}_{m+1}y(\lambda,m)=y(\lambda,m+1).$$

Multiply equation (3) by H_m:

$$(H_mH^{\dagger}_m)(H_my)=[\lambda-L(m)](H_my),$$

Comparing with equation (2),

$$H_my(\lambda,m)=y(\lambda,m-1).$$

H_m and $H^{\dagger}_m$ are ladder operators with respect to m, H_m going down and $H^{\dagger}_m$ going up.

Theorem. H_m and $H_m^\dagger$ are adjoint operators (for example, Hermitian) if

$$\int_a^b \phi(H_m\psi)\,dx = \int_a^b (H_m^\dagger\phi)\psi\,dx,$$

where $\phi\psi$ vanishes at the end points and the integrand is continuous in (a,b).

Proof. Already done.

Inner product:

$$(\phi,\psi) = \int_a^b \phi^*\psi\,dx.$$

Norm:

$$\|\phi\|^2 = (\phi,\phi).$$

Theorem. If $y(\lambda,m)$ is L^2 over (a,b) and $L(m)$ is an increasing function of $m>0$, then the raising operator $H^\dagger$ produces an L^2 function which vanishes at the end points. If $L(m)$ is a decreasing function of $m>0$, the lowering operator H produces an L^2 function which vanishes at the end points.

[Just accept as the proof has to be done for every particular case – not the L^2 part but the vanishing.]

Theorem. When $L(m)$ is an increasing function of the integer m for $0<m\le M$ ($\le\infty$), and $\lambda\le\max[L(m),L(m+1)]$, a necessary condition for L^2 solutions is that

$$\lambda = \lambda_l = L(l+1), \qquad m=0,1,\ldots,l.$$

Proof. Assume that $y(\lambda,m)$ is L^2. Then

$$y(\lambda,m+1) = H_{m+1}^\dagger y(\lambda,m)$$

is also L^2 and vanishes at the endpoints (see previous theorem).

$$\begin{aligned}\|y(\lambda,m+1)\|^2 &= \Big(H_{m+1}^\dagger y(\lambda,m), H_{m+1}^\dagger y(\lambda,m)\Big)\\ &= \Big(y(\lambda,m), H_{m+1}H_{m+1}^\dagger y(\lambda,m)\Big)\\ &= [\lambda - L(m+1)]\|y(\lambda,m)\|^2.\end{aligned}$$

Likewise,

$$\begin{aligned}\|y(\lambda,m+2)\|^2 &= [\lambda - L(m+2)]\|y(\lambda,m+1)\|^2\\ &= [\lambda - L(m+2)][\lambda - L(m+1)]\|y(\lambda,m)\|^2.\end{aligned}$$

L is an increasing function of m, and so for some integer l we will have $\lambda - L(l+1) < 0$, that is, $\|y(\lambda,l+1)\|^2 < 0$, which is against the rules, so we have

$$\lambda = L(l+1), \qquad H_{l+1}^\dagger y(\lambda,l) = 0.$$

This fixes λ in terms of l, the maximum value of m. All other m are less than l, and the eigenvalue is $L(l+1)-L(m+1)$.

Theorem. If $L(m)$ is a decreasing function of m for $0<m\le M$ ($\le\infty$), and $\lambda<L(\infty)$, a necessary condition for the existence of an L^2 solution is that

$$\lambda = \lambda_l = L(l), \qquad m = l, l+1, l+2, \ldots.$$

Proof. Assume that $y(\lambda, m)$ is L^2:

$$\begin{aligned}\|y(\lambda, m-1)\|^2 &= \Big(H_m y(\lambda, m), H_m y(\lambda, m)\Big)\\ &= \Big(y(\lambda, m), H_m^\dagger H_m y(\lambda, m)\Big)\\ &= [\lambda - L(m)]\|y(\lambda, m)\|^2,\end{aligned}$$

Likewise,

$$\|y(\lambda, m-2)\|^2 = [\lambda - L(m-1)][\lambda - L(m)]\|y(\lambda, m)\|^2.$$

Continue until there is an integer l such that $\lambda - L(l) < 0$. This gives a contradiction, so

$$\lambda = L(l), \qquad \|y(\lambda, l-1)\| = 0.$$

That is, $y(\lambda, l-1) = 0$, or $H_l y(\lambda, l) = 0$. l is the lowest value of m, and $m = l, l+1, l+2, \ldots$.

In the above we have taken $m > 0$. This is not necessary. m may start at some m_0, not necessarily an integer. Then $|l - m|$ must be an integer. We now write the solutions as $y_l^m(x)$, or $Y_l^m(x)$ when normalised.

Suppose $Y_l^m(x)$ is normalised, that is,

$$\|Y_l^m(x)\|^2 = \int_a^b Y_l^{m*} Y_l^m \, dx = 1.$$

Then

$$y_l^{m+1} = H_{m+1}^\dagger Y_l^m,$$

so that

$$\begin{aligned}\|y_l^{m+1}\|^2 &= \|H_{m+1}^\dagger Y_l^m\|^2\\ &= \Big(H_{m+1}^\dagger Y_l^m, H_{m+1}^\dagger Y_l^m\Big)\\ &= \Big(Y_l^m, H_{m+1} H_{m+1}^\dagger Y_l^m\Big)\\ &= [L(l+1) - L(m+1)]\|Y_l^m\|^2\\ &= L(l+1) - L(m+1).\end{aligned}$$

That is, y_l^{m+1} is not normalised. Likewise,

$$y_l^{m-1} = H_m Y_l^m,$$

and

$$\|y_l^{m-1}\|^2 = [L(l) - L(m)]\|Y_l^m\|^2 = L(l) - L(m),$$

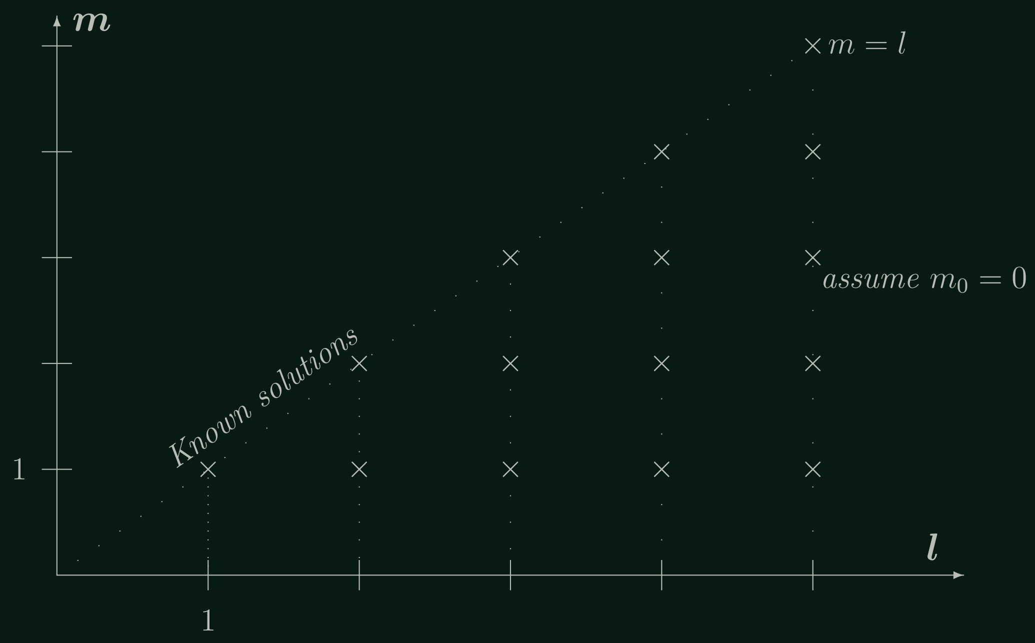
that is, y_l^{m-1} is not normalised. Overcome the problem by defining

$$\begin{aligned}\mathcal{H}_l^{\dagger, m+1} &= [L(l+1) - L(m+1)]^{-1/2} H_{m+1}^\dagger,\\ \mathcal{H}_l^m &= [L(l) - L(m)]^{-1/2} H_m.\end{aligned}$$

If we calculate y_l^l from either $H_{l+1}^\dagger y_l^l = 0$ (up) or $H_l y_l^l = 0$ (down), and normalise to Y_l^l, then the Y_l^m generated by $\mathcal{H}^\dagger$ and $\mathcal{H}$ are normalised automatically. The differential equation becomes

$$\mathcal{H}_l^{m+1}\mathcal{H}_l^{\dagger,m+1}Y_l^m = Y_l^m, \qquad \mathcal{H}_l^{\dagger,m}\mathcal{H}_l^m Y_l^m = Y_l^m.$$

Chart for class 1 solutions: $L(m)$ is increasing.



Basic solution y_l^l comes from

$$H_{l+1}^\dagger y_l^l = 0,$$

which is a first-order differential equation:

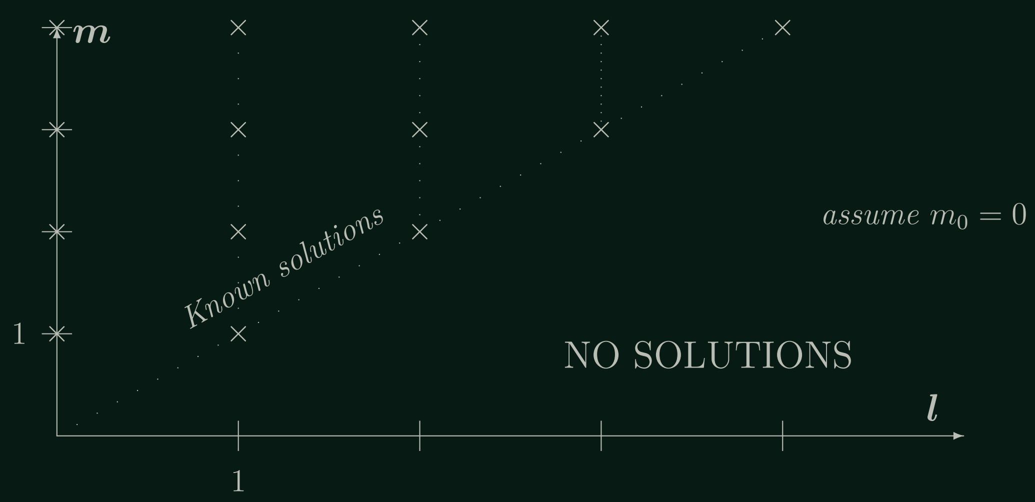
$$\left[k(x,l+1) - \frac{d}{dx}\right] y_l^l = 0.$$

$$y_l^l = C\exp\left(\int k(x,l+1)\,dx\right).$$

C is found from requiring $\|y_l^l\| = 1$. Obtain Y_l^m from

$$\mathcal{H}_l^m Y_l^m = Y_l^{m-1}, \qquad m = l, l-1, \ldots, 0.$$

Chart for class II solutions: $L(m)$ is a decreasing function.



Basic solutions Y_l^l come from

$$H_l y_l^l = 0$$

that is,

$$\left[k(x,l) + \frac{d}{dx}\right] Y_l^l = 0,$$

so

$$y_l^l = C \exp\left(-\int k(x,l)\,dx\right),$$

C is determined by the normalisation condition $\|Y_l^l\| = 1$.
Factorisation types:

$$[-r(x,m) - D^2]y = \lambda y.$$

Seek

$$H_m = k(x,m) + D, \qquad H_m^\dagger = k(x,m) - D,$$

such that

$$H_{m+1}H_{m+1}^\dagger y = [\lambda - L(m+1)]y,$$
$$H_m^\dagger H_m y = [\lambda - L(m)]y.$$

that is,

$$H_{m+1}H_{m+1}^\dagger + L(m+1) = -r(x,m) - D^2,$$

and

$$H_m^\dagger H_m + L(m) = -r(x,m) - D^2,$$

which is

$$k^2(x,m+1)+k'(x,m+1)+L(m+1)=-r(x,m),$$
$$k^2(x,m)-k'(x,m)+L(m)=-r(x,m).$$

Subtracting,

$$k^2(x,m+1)-k^2(x,m)+k'(x,m+1)+k'(x,m)=L(m)-L(m+1).$$

What k's and L's?

1. Let $k(x,m)=f(m)$. Then $L(m)=-f^2(m)$ and the differential equation is

$$y''+\lambda y=0,$$

which is a bit simple.

2. Let

$$k(x,m)=k_0(x)+mk_1(x).$$

Substituting

$$\begin{aligned}(m+1)^2(k_1^2+k_1')+2(m+1)(k_0k_1+k_0')\\ -m^2(k_1^2+k_1')-2m(k_0k_1+k_0')=L(m)-L(m+1).\end{aligned}$$

that is,

$$L(m)=-\left[m^2(k_1^2+k_1')+2m(k_0k_1+k_0')+f(m,x)\right],$$

where $f(m+1,x)=f(m,x)$, that is, $f(m,x)$ has period one in m. Since we are only interested in integer increments in m, take $f(m,x)=f(x)$. Separate by coefficients of powers of m:

$$k_1^2+k_1'=-a^2$$

and

$$k_0k_1+k_0'=\begin{cases}-ca, & a\neq 0,\\ b, & a=0.\end{cases}$$

For m^0, $f(x)$ is constant.

Since $f(x)$ is constant it just shifts $L(m)$ by a constant amount for any m; the constant may take any value and is usually zero.

Case $a\neq 0$,

$$\begin{aligned}&\text{A:}\quad k_1=a\cot a(x+p),\quad k_0=ca\cot a(x+p)+\frac{d}{\sin a(x+p)},\\ &\text{B:}\quad k_1=ia,\quad k_0=cia+de^{-iax}\quad\text{(a particular solution)},\end{aligned}$$

Case $a=0$,

$$\begin{aligned}&\text{C:}\quad k_1=\frac{1}{x},\quad k_0=\frac{1}{2}bx+\frac{d}{x},\\ &\text{D:}\quad k_1=0,\quad k_0=bx+d\quad\text{(a particular solution)}.\end{aligned}$$

3. Try

$$k(x,m) = \sum_{i=0}^{n} k_i(x) m^i, \qquad n < \infty.$$

leads to nothing new.

4. Try

$$k(x,m) = \frac{k_{-1}(x)}{m} + k_0(x) + m k_1(x)$$

leads to

$$\text{E:} \quad k_1 = a \cot a(x+p), \quad k_0 = 0, \quad k_{-1} = q,$$

$$\text{F:} \quad k_1 = \frac{1}{x}, \quad k_0 = 0, \quad k_{-1} = q.$$

For E and F,

$$L(m) = a^2 m^2 - \frac{q^2}{m^2},$$

with $a = 0$ in F. Trying

$$k(x,m) = \sum_{i=-n}^{1} k_i(x) m^i, \qquad n < \infty,$$

Nothing new!

Examples:

Type A:

$$r(x,m) = \frac{-a^2(m+c)(m+c+1) + d^2 + 2ad\left(m+c+\frac{1}{2}\right)\cos a(x+p)}{\sin^2 a(x+p)}.$$

$$k = (m+c) a \cot a(x+p) + \frac{d}{\sin a(x+p)},$$

$$L(m) = a^2(m+c)^2.$$

Associated spherical harmonics

The "angle" operator in spherical polars with radial symmetry in the field is

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{dP}{d\theta}\right) + \left[\lambda - \frac{m^2}{\sin^2\theta}\right] P = 0.$$

where $0 \leq \theta \leq \pi$.

To put into standard form let

$$Y = (\sin\theta)^{1/2} P,$$

which gives

$$Y'' - \frac{m^2 - \frac{1}{4}}{\sin^2\theta} Y + \left(\lambda + \frac{1}{4}\right) Y = 0.$$

Comparing with the standard A form for $r(x, m)$,

$$a = 1, \qquad p = 0, \qquad d = 0, \qquad c = -\frac{1}{2},$$

and

$$\lambda' = \lambda + \frac{1}{4},$$

so that

$$k(\theta, m) = \left(m - \frac{1}{2}\right) \cot\theta, \qquad L(m) = \left(m - \frac{1}{2}\right)^2.$$

Since $L(m)$ is an increasing function of m for $2m > 0$, this is a class 1 problem, that is, there exists a value $m = l$ such that

$$\lambda' = L(l+1) = \left(l + \frac{1}{2}\right)^2,$$

so

$$\lambda = \lambda' - \frac{1}{4} = l(l+1), \qquad l = 0, 1, 2, \ldots, \quad l \geq m.$$

Eigenfunctions:

$$H^\dagger_{l+1} y^l_l = 0.$$

$$\left[k(\theta, l+1) - \frac{d}{d\theta}\right] y^l_l = 0.$$

$$\left[\left(l + \frac{1}{2}\right) \cot\theta - \frac{d}{d\theta}\right] Y^l_l = 0,$$

$$\left(l + \frac{1}{2}\right) \frac{\cos\theta}{\sin\theta}\, d\theta = \frac{dY^l_l}{Y^l_l}.$$

$$Y^l_l(\theta) = K \sin^{l+1/2}\theta.$$

To find the normalised eigenfunction Y^l_l,

$$\begin{aligned}
K^{-2} &= \int_0^\pi (\sin\theta)^{l+1/2} (\sin\theta)^{l+1/2}\, d\theta \\
&= \int_0^\pi (\sin\theta)^{2l+1}\, d\theta \\
&= \frac{2}{1}\frac{2}{3}\frac{4}{5} \cdots \frac{2l}{2l+1} \\
&= \frac{2}{1}\frac{2}{2}\frac{2}{3}\frac{4}{4}\frac{4}{5} \cdots \frac{2l}{2l}\frac{2l}{2l+1} \\
&= \frac{2^{2l+1} l!\, l!}{(2l+1)!}.
\end{aligned}$$

$$Y_l^l(\theta) = \left[\frac{(2l+1)!}{2^{2l+1}l!\,l!}\right]^{1/2} \sin^{l+1/2}\theta.$$

Also,

$$Y_l^{m-1} = \frac{\left[\left(m-\frac{1}{2}\right)\cot\theta + \dfrac{d}{d\theta}\right] Y_l^m}{\sqrt{(l+m)(l-m+1)}},$$

and

$$Y_l^{m+1} = \frac{\left[\left(m+\frac{1}{2}\right)\cot\theta - \dfrac{d}{d\theta}\right] Y_l^m}{\sqrt{(l+m+1)(l-m)}}.$$

Note that replacing m by $-m$ in the original differential equation does not change the problem. Hence solutions for $-m$, with $m = 0, 1, \ldots, l$, exist and

$$Y_l^{-m} = (-1)^m Y_l^m.$$

Associated spherical harmonics as a class 2 problem. Go back to

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP_l^m}{d\theta}\right) + \left[l(l+1) - \frac{m^2}{\sin^2\theta}\right] P_l^m = 0,$$

and replace m^2 by λ and λ by $l(l+1)$, so that

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) - \frac{\lambda}{\sin^2\theta}P + l(l+1)P = 0.$$

Normal form: let

$$z = \log\tan\frac{\theta}{2}.$$

and let $P(\theta) \mapsto Q(z)$ to obtain

$$\frac{d^2Q}{dz^2} + \frac{l(l+1)}{\cosh^2 z}Q + \lambda Q = 0,$$

Comparing with standard form, take

$$d = 0, \qquad p = \frac{\pi}{2}, \qquad a = i, \qquad c = 0,$$

and replace x, m with z, l. Then

$$k(z,l) = l\tanh z, \qquad L(l) = -l^2.$$

which is patently class 2.

$$\frac{d^2Q}{dz^2} + \frac{l(l+1)}{\cosh^2 z}Q + \lambda Q = 0.$$

The bottom of the ladder is reached when l is some value m:

$$\lambda_m = L(m) = -m^2, \qquad l - m = 0, 1, 2, \ldots.$$

Then

$$\begin{aligned} Q_m^m &= C \exp\left(-\int m \tanh z \, dz\right) \\ &= C \cosh^{-m} z. \end{aligned}$$

Also,

$$\begin{aligned} \|Q_m^m\|^2 &= \int_{-\infty}^{\infty} C^2 \cosh^{-2m} z \, dz \\ &= 2C^2 \int_0^{\infty} \cosh^{-2m} z \, dz \\ &= 2C^2 4^{m-1} B(m, m). \qquad (*) \end{aligned}$$

$$I = \int_0^{\infty} \operatorname{sech}^{2m} z \, dz.$$

(i)

$$\begin{aligned} I &= \int_0^{\infty} \operatorname{sech}^{2m-2} z \operatorname{sech}^2 z \, dz \\ &= \int_0^1 (1 - \eta^2)^{m-1} \, d\eta \\ &= \sum_{i=0}^{m-1} (-1)^i \binom{m-1}{i} \int_0^1 \eta^{2i} \, d\eta \\ &= \sum_{i=0}^{m-1} (-1)^i \frac{1}{2i+1} \binom{m-1}{i}. \end{aligned}$$

(ii)

$$\begin{aligned} I &= \int_0^{\pi/2} \sin^{2m-1} \theta \, d\theta \\ &= \int_0^{\pi/2} \sin^{2m-2} \theta \sin \theta \, d\theta \\ &= -\cos \theta \sin^{2m-2} \theta \Big|_0^{\pi/2} + (2m-2) \int_0^{\pi/2} \sin^{2m-3} \theta \cos^2 \theta \, d\theta \\ &= (2m-2) \int_0^{\pi/2} \sin^{2m-3} \theta \, d\theta - (2m-2) \int_0^{\pi/2} \sin^{2m-1} \theta \, d\theta. \end{aligned}$$

Hence

$$I = \frac{2m-2}{2m-1} \frac{2m-4}{2m-3} \frac{2m-6}{2m-5} \cdots \frac{4}{3} \frac{2}{3} = \frac{\{2^{m-1}(m-1)!\}^2}{(2m-1)!}.$$

From $(*)$, where $B(m, m)$ is the Beta function,

$$1 = C^2 2^{2m-1} B(m, m),$$

$$C^{-2} = 2^{2m-1}\frac{\Gamma(m)\Gamma(m)}{\Gamma(m+m)}$$
$$= 2^{2m-1}\frac{(m-1)!(m-1)!}{(2m-1)!}.$$

Beta function:

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt.$$

$$\int_0^1 (1-\eta^2)^{m-1}\,d\eta = \int_0^1 (1-\eta)^{m-1}(1+\eta)^{m-1}\,d\eta.$$

Plan I:

$$1-\eta = 2\xi,$$

$$1+\eta = 2-(1-\eta) = 2-2\xi.$$

I becomes

$$I = 2\int_{1/2}^1 (2-2\xi)^{m-1}(2\xi)^{m-1}\,d\xi$$
$$= 2^{2m-1}\int_{1/2}^1 \xi^{m-1}(1-\xi)^{m-1}\,d\xi.$$

(symmetric about $\xi = \frac{1}{2}$ since both powers are $m-1$)

$$I = 2^{2m-2}\int_0^1 \xi^{m-1}(1-\xi)^{m-1}\,d\xi$$
$$= 2^{2m-2}B(m,m).$$

$$C = \frac{[2(2m-1)!]^{1/2}}{2^m(m-1)!},$$

so

$$Q_m^m(z) = \frac{[2(2m-1)!]^{1/2}}{2^m(m-1)!}\cosh^{-m} z.$$

Note that

$$Q_l^m = \frac{l\tanh z - \dfrac{d}{dz}}{\sqrt{(l-m)(l+m)}}Q_{l-1}^m, \qquad l-1 \geq m.$$

In the original coordinate,

$$P_l^m = \frac{-l\cos\theta - \sin\theta\dfrac{d}{d\theta}}{\sqrt{(l-m)(l+m)}}P_{l-1}^m,$$

and

$$P_{l-1}^m = \frac{-l\cos\theta + \sin\theta \dfrac{d}{d\theta}}{\sqrt{(l-m)(l+m)}} P_l^m.$$

Eigenfunctions of the spherical harmonics associated with $SO(3)$

Rotation about Oz:

$$\bar{x} = x\cos\theta + y\sin\theta, \qquad \bar{y} = -x\sin\theta + y\cos\theta, \qquad \bar{z} = z.$$

For some angle θ. an infinitesimal angle $d\theta$,

$$\bar{x} = x + y\,d\theta, \qquad \bar{y} = y - x\,d\theta, \qquad \bar{z} = z.$$

Seek a generator of the infinitesimal transformation.
In general, under an infinitesimal transformation generated by G, $f(x, y, z)$ goes to

$$\bar{f} = (1 + \varepsilon G)f, \qquad G = \eta_1 \frac{\partial}{\partial x} + \eta_2 \frac{\partial}{\partial y} + \eta_3 \frac{\partial}{\partial z}.$$

For $f = x$,

$$\bar{x} = x + \varepsilon\eta_1.$$

For $f = y$,

$$\bar{y} = y + \varepsilon\eta_2.$$

For $f = z$,

$$\bar{z} = z + \varepsilon\eta_3.$$

Comparing,

$$\eta_1 = y, \qquad \eta_2 = -x, \qquad \eta_3 = 0,$$

$$G_1 = y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}.$$

Similarly, for rotations about Ox,

$$G_2 = z\frac{\partial}{\partial y} - y\frac{\partial}{\partial z},$$

and about Oy,

$$G_3 = x\frac{\partial}{\partial z} - z\frac{\partial}{\partial x}.$$

$$[G_1, G_2] = G_3,$$
$$[G_2, G_3] = G_1,$$
$$[G_3, G_1] = G_2,$$

$$[G_i, G_j] = \varepsilon_{ijk} G_k,$$

that is, a standard representation of $SO(3)$.

Interested in invariance under rotation, that is, spherical symmetry.

So convert to spherical polars:

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta.$$

$$\begin{aligned}
\frac{\partial}{\partial r} &= \sin\theta\cos\phi\frac{\partial}{\partial x} + \sin\theta\sin\phi\frac{\partial}{\partial y} + \cos\theta\frac{\partial}{\partial z}, \\
\frac{1}{r}\frac{\partial}{\partial\theta} &= \cos\theta\cos\phi\frac{\partial}{\partial x} + \cos\theta\sin\phi\frac{\partial}{\partial y} - \sin\theta\frac{\partial}{\partial z}, \\
\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi} &= -\sin\phi\frac{\partial}{\partial x} + \cos\phi\frac{\partial}{\partial y}.
\end{aligned}$$

$$\begin{pmatrix} \partial_x \\ \partial_y \\ \partial_z \end{pmatrix} = \begin{pmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{pmatrix}^{-1} \begin{pmatrix} \partial_r \\ r^{-1}\partial_\theta \\ (r\sin\theta)^{-1}\partial_\phi \end{pmatrix}.$$

which gives

$$\begin{aligned}
G_1 &= -\frac{\partial}{\partial\phi}, \\
G_2 &= \sin\phi\frac{\partial}{\partial\theta} + \cos\phi\cot\theta\frac{\partial}{\partial\phi}, \\
G_3 &= -\cos\phi\frac{\partial}{\partial\theta} + \sin\phi\cot\theta\frac{\partial}{\partial\phi}.
\end{aligned}$$

Define:

$$\begin{aligned}
L_\pm &= i(G_2 \pm iG_3) \\
&= i\left[\sin\phi\frac{\partial}{\partial\theta} + \cos\phi\cot\theta\frac{\partial}{\partial\phi}\right] \\
&\quad \pm \left[-\cos\phi\frac{\partial}{\partial\theta} + \sin\phi\cot\theta\frac{\partial}{\partial\phi}\right] \\
&= \pm(\cos\phi \pm i\sin\phi)\frac{\partial}{\partial\theta} + i(\cos\phi \pm i\sin\phi)\cot\theta\frac{\partial}{\partial\phi} \\
&= e^{\pm i\phi}\left(i\cot\theta\frac{\partial}{\partial\phi} \pm \frac{\partial}{\partial\theta}\right),
\end{aligned}$$

$$\begin{aligned}
[L_+, L_-] &= [i(G_2 + iG_3), i(G_2 - iG_3)] \\
&= -i[G_3, G_2] + i[G_2, G_3] \\
&= 2i[G_2, G_3] \\
&= 2iG_1 \\
&= -2i\frac{\partial}{\partial\phi}.
\end{aligned}$$

Define:

$$L_3 = -i\frac{\partial}{\partial\phi} = iG_1.$$

For a spherically symmetric problem involving the Laplacian, the differential equation is of the form

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r} + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + f(r)\right]\psi = 0.$$

Let

$$\psi = R(r)\Phi(\theta,\phi).$$

Then

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + r^2 f(r) = -\frac{1}{\Phi}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]\Phi = \lambda,$$

which gives rise to the eigenvalue problem

$$-\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]\Phi = \lambda\Phi.$$

Observe that

$$L^2 = -(G_1^2+G_2^2+G_3^2) = -\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right].$$

Looking at the problem,

$$L^2\psi = \lambda\psi.$$

Also,

$$\begin{aligned}
[L^2, L_3] &= [-G_1^2 - G_2^2 - G_3^2, iG_1] \\
&= -i[G_2^2, G_1] - i[G_3^2, G_1] \\
&= -iG_2[G_2, G_1] - i[G_2, G_1]G_2 \\
&\quad - iG_3[G_3, G_1] - i[G_3, G_1]G_3 \\
&= iG_2G_3 + iG_3G_2 - iG_3G_2 - iG_2G_3 \\
&= 0,
\end{aligned}$$

This means that L^2 and L_3 have simultaneous eigenfunctions.
Let ψ be the eigenfunction of L^2. Then

$$\begin{aligned}
L_3L^2\psi &= ([L_3, L^2] + L^2L_3)\psi \\
&= L^2L_3\psi,
\end{aligned}$$

so $L^2(L_3\psi) = \lambda(L_3\psi)$. Moreover, ψ and $L_3\psi$ are eigenfunctions to the same eigenvalue λ.
Write

$$L^2\psi_{\lambda,\mu} = \lambda\psi_{\lambda,\mu}, \qquad L_3\psi_{\lambda,\mu} = \mu\psi_{\lambda,\mu}.$$

Consider

$$
\begin{aligned}
L_3(L_\pm \psi_{\lambda,\mu}) &= \Big([L_3, L_\pm] + L_\pm L_3\Big)\psi_{\lambda,\mu} \\
&= (\pm L_\pm + \mu L_\pm)\psi_{\lambda,\mu} \\
&= (\mu \pm 1) L_\pm \psi_{\lambda,\mu},
\end{aligned}
$$

that is, $L_\pm$ are ladder operators for μ:

$$
L_\pm \psi_{\lambda,\mu} = (\text{constant})\psi_{\lambda,\mu\pm 1}.
$$

Now

$$
\begin{aligned}
(L_+L_-)^\dagger &= (L_-)^\dagger (L_+)^\dagger = L_+L_-, \\
(L_-L_+)^\dagger &= (L_+)^\dagger (L_-)^\dagger = L_-L_+,
\end{aligned}
$$

that is, L_+L_- and L_-L_+ are both Hermitian operators. Just as for matrices of the form $AA^\dagger \sim A^\dagger A$, the eigenvalues are real and non-negative, having the form $\bar{\lambda}\lambda$, that is, $|\lambda|^2$, so L_+L_- and L_-L_+ have non-negative eigenvalues.

Now

$$
\begin{aligned}
L_+L_- &= i(G_2 + iG_3)i(G_2 - iG_3) \\
&= -G_2^2 - G_3^2 + i[G_2, G_3] \\
&= -(G_1^2 + G_2^2 + G_3^2) + G_1^2 + iG_1 \\
&= L^2 - L_3^2 + L_3,
\end{aligned}
$$

and

$$
\begin{aligned}
L_-L_+ &= i(G_2 - iG_3)i(G_2 + iG_3) \\
&= L^2 - L_3^2 - L_3.
\end{aligned}
$$

Hence

$$
\begin{aligned}
L_+L_-\psi_{\lambda,\mu} &= (\lambda - \mu^2 + \mu)\psi_{\lambda,\mu}, \\
L_-L_+\psi_{\lambda,\mu} &= (\lambda - \mu^2 - \mu)\psi_{\lambda,\mu}.
\end{aligned}
$$

Since L_+L_- and L_-L_+ have non-negative eigenvalues,

$$
\lambda - \mu^2 + \mu \geq 0, \qquad \lambda - \mu^2 - \mu \geq 0.
$$

Since L_- is a lowering operator, $\lambda - \mu^2 + \mu$ will eventually become negative unless there exists μ' such that

$$
L_-\psi_{\lambda,\mu'} = 0
$$

and so

$$
\lambda - \mu'^2 + \mu' = 0.
$$

Likewise, L_+ is a raising operator and $\lambda - \mu^2 - \mu$ will become negative unless there exists a μ'' such that

$$
L_+\psi_{\lambda,\mu''} = 0
$$

and

$$\lambda - \mu''^2 - \mu'' = 0.$$

L_+ raises (L_- lowers) μ by integer steps, and so

$$\mu'' - \mu' = \text{integer} = n \geq 0.$$

At μ',

$$\lambda = \mu'^2 - \mu',$$

and at μ'',

$$\lambda = \mu''^2 + \mu''.$$

and the value of λ is the same, so

$$\begin{aligned} 0 &= (\mu'' - \mu')(\mu'' + \mu') + (\mu'' + \mu') \\ &= (\mu'' + \mu')(\mu'' - \mu' + 1) \\ &= (\mu'' + \mu')(n+1), \end{aligned}$$

so $\mu' = -\mu''$, and so $2\mu'' = n$.
Let

$$\mu'' = l, \qquad \mu' = -l.$$

$2l$ is an integer and

$$\lambda = \mu'^2 - \mu' = l(l+1).$$

Because of the integer steps in μ, 2μ is an integer; now replace μ by m.
To find the eigenfunction,

$$L_3 = -i\partial_\phi,$$

$$L_3\psi_{l,m} = m\psi_{l,m},$$

$$-i\frac{\partial\psi_{l,m}}{\partial\phi} = m\psi_{l,m},$$

so

$$\psi_{l,m} = e^{im\phi} f(\theta).$$

If ψ is to be a single valued function of ϕ, then the half-integer values must be excluded.
Now $L_+\psi_{l,l} = 0$, so

$$\begin{aligned} 0 &= L_+\psi_{l,l} \\ &= e^{i\phi}\left(i\cot\theta\frac{\partial}{\partial\phi} + \frac{\partial}{\partial\theta}\right)e^{il\phi}f(\theta) \\ &= e^{i(l+1)\phi}\left[-l\cot\theta f + f'\right]. \end{aligned}$$

Hence

$$f = K \sin^l \theta.$$

The norm is

$$\begin{aligned} \|\psi_{l,l}\|^2 &= \int_0^\pi \sin\theta\, d\theta \int_0^{2\pi} d\phi\, \psi_{l,l}^* \psi_{l,l} \\ &= 2\pi |K|^2 \int_0^\pi \sin^{2l+1}\theta\, d\theta \\ &= 2\pi |K|^2 \frac{2^{2l+1} l!\, l!}{(2l+1)!}. \end{aligned}$$

$$\psi_{l,l} = \frac{1}{2^{l+1} l!} \left[\frac{(2l+1)!}{\pi} \right]^{1/2} e^{il\phi} \sin^l \theta.$$

In general,

$$\psi_{l,m-1} = \frac{1}{\sqrt{l(l+1) - m(m-1)}} L_- \psi_{l,m}.$$

or

$$\psi_{l,m-1} = \ell_- \psi_{l,m},$$

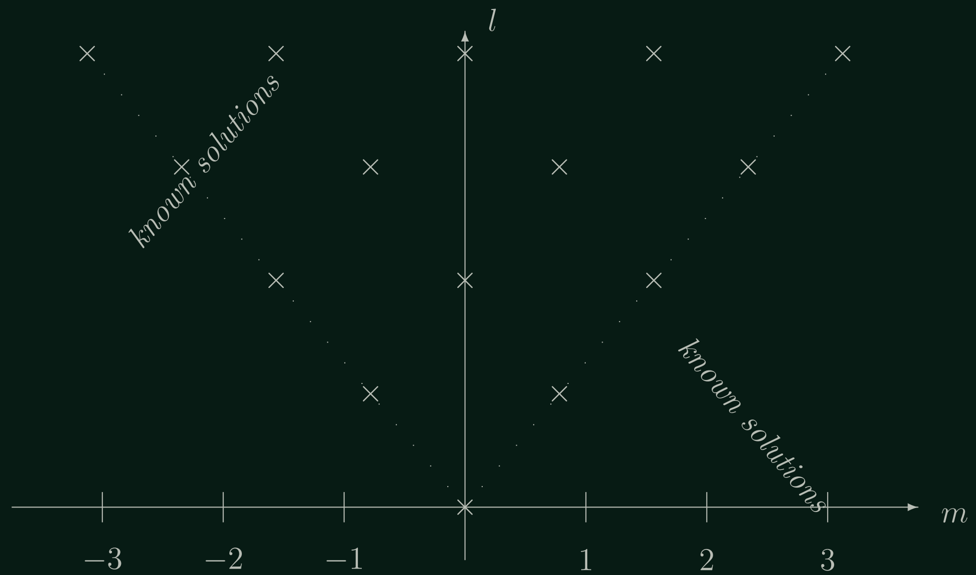
where

$$\ell_- = [l(l+1) - m(m-1)]^{-1/2} L_-.$$

For example,

$$\psi_{l,l-1} = -\frac{1}{2^l (l-1)!} \left[\frac{(2l+1)!}{2\pi l} \right]^{1/2} e^{i(l-1)\phi} \cos\theta \sin^{l-1}\theta.$$

And so on.



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